CHAPTER

6

Redox Reactions, Equivalent Concept, Normality

For NSEJS 2025

GENERAL INTRODUCTION OF CHEMISTRY & CHARACTERISTICS OF CHEMICAL REACTIONS, TYPE OF CHEMICAL REACTIONS (REDOX REACTIONS)

INTRODUCTION

Chemistry is defined as that branch of science which deals with the composition and properties of matter and the changes that matter undergoes by various interactions.

A chemical compound is formed as a result of a chemical change and in this process different types of energies such as heat, electrical energy, radiation etc. are either absorbed or evolved. The total mass of the substance remains the same throughout the chemical change.

CHEMICAL REACTION AND EQUATION:

When a chemical change occurs, a chemical action is said to have taken place. A chemical change or chemical action is represented by a chemical equation. The matter undergoing change is known as reactant and new chemical component formed is known as product.

(a) Characteristics of a Chemical Reaction:

When we heat sugar crystals they melt and on further heating they give steamy vapour, leaving behind brownish black mass. On cooling no sugar crystals appear. Thus change which takes place on heating sugar is a chemical change and the process which brings about this chemical change is called chemical reaction.

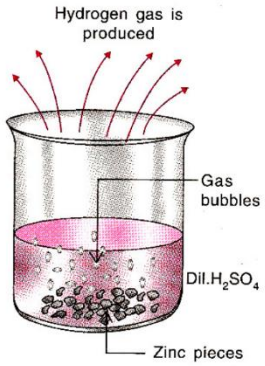
- In this reaction the substances which take part in bringing about chemical change are called reactants.
- The substances which are produced as a result of chemical change are called products.
- These reactions involve breaking and making of chemical bonds.
- Product(s) of the reaction is/are new substances with new name(s) and chemical formulae.
- It is often difficult to reverse a chemical reaction.
- Properties of products formed during a chemical reaction are different from those of the reactants.
- Apart from heat other forms of energies are light and electricity which are also used in carrying out chemical changes.

In all chemical reactions, the transformation from reactants to products is accompanied by various characteristics, which are-

(i) Evolution of gas:

Some chemical reactions take place with the evolution of a gas.

Some reactions in which a gas is evolved are described below:



Reaction between a metal such as zinc, magnesium, or iron and dilute sulphuric acid produces hydrogen gas.

$Zn\ (s)$	+	$H_2SO_4\ (dil)$	$\longrightarrow$	$ZnSO_4\ (aq)$	+	$H_2\ (g)$
zinc		sulphuric acid		zinc sulphate		hydrogen gas

Reaction between iron sulphide and dilute sulphuric acid produces hydrogen sulphide gas.

$FeS\ (s)$	+	$H_2SO_4(dil)$	$\longrightarrow$	$FeSO_4\ (aq) + H_2S\ (g)$
iron sulphide				ferrous sulphate

Heating a mixture of potassium chlorate ($KClO_3$) with manganese dioxide (MnO_2) gives oxygen gas.

$2KClO_3\ (s)$	$\longrightarrow$	$2KCl\ (s) + 3O_2(g)$
Potassium chlorate		potassium chloride oxygen gas

(MnO_2 is used as a catalyst in this reaction.)

This reaction is used for the preparation of oxygen in the laboratory.

(ii) Change of colour: Certain chemical reactions are characterised by the change in colour of reacting substances.

- When red lead oxide is heated strongly it forms yellow coloured lead monoxide and gives off oxygen gas.

$2Pb_3O_4\ (s)$	$\xrightarrow{Heat}$	$6PbO(s)$	+	$O_2\ (g)$
Lead Oxide		Lead monoxide		
(Red)		(Yellow)		

- When copper carbonate (green) is heated strongly it leaves behind a black residue.

$CuCO_3\ (s)$	$\xrightarrow{Heat}$	$CuO(s)$	+	$CO_2\ (g)$
copper Carbonate (Green)	(Black)	Copper oxide		Carbon dioxide

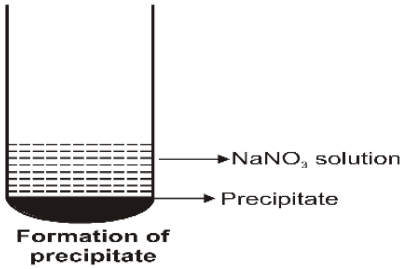
Other such examples are –

$2Pb(NO_3)_2\ (s)$	$\xrightarrow{Heat}$	$2PbO(s)$	+	$4NO_2\ (g)$	+	$O_2\ (g)$
Lead (II) Nitrate		Lead (II) oxide		Nitrogen dioxide		
(White)		(Yellow)		(Brown)		

$C_{12}H_{22}O_{11}\ (s)$	$\xrightarrow{Heat}$	$12C(s)$	+	$11H_2O\ (g)$
White sugar		Carbon black		Water

(iii) Formation of precipitate: Some chemical reactions are characterized by the formation of precipitate (an insoluble substance), when the solutions of two soluble chemical compounds are mixed together.

- When silver nitrate solution is mixed with a solution of sodium chloride, a white precipitate of silver chloride is formed.



$AgNO_3(aq) + NaCl\ (aq) \longrightarrow NaNO(aq) + AgCl\ (s)$

Silver nitrate	Sodium chloride (Colourless)	Sodium nitrate (Colourless)	Silver chloride (White precipitate)

$FeSO_4(aq) + 2NaOH\ (aq) \longrightarrow Na_2SO_4(aq) + Fe(OH)_2(s)$

Ferrous sulphate	Sodium hydroxide	Sodium sulphate	Ferrous hydroxide (Green precipitate)

(iv) Energy changes: All chemical reactions proceed either with the absorption or release of energy.
On the basis of energy changes, there are two types of reactions:

(A) Endothermic reaction: A chemical reaction which is accompanied by the absorption of energy is called an endothermic reaction.

$C\ (s) + 2S\ (s) \xrightarrow{Heat} CS_2\ (\ell)$

- Light energy is essential for biochemical reaction (photosynthesis) by which green plants prepare their food from carbon dioxide & water.

(B) Exothermic reaction: A chemical reaction which is accompanied by the release of energy is called exothermic reaction.

- When magnesium wire is heated from its tip in a bunsen flame, it catches fire and burns with a dazzling white flame with release of heat and light energy.

$2Mg\ (s) + O_2\ (g) \xrightarrow{Heat} 2MgO\ (s) + Energy$

- When quick lime (calcium oxide) is placed in water, the water becomes very hot and sometimes starts boiling. It is because of release of heat energy during the reaction.

(v) Change of state: Some chemical reactions are characterized by a change in state - i.e. solid, liquid or gas.

- Two volumes of hydrogen gas react with one volume of oxygen gas to form water.

$2H_2\ (g) + O_2\ (g) \longrightarrow 2H_2O\ (\ell)$

or when electric current is passed through water it splits into its elements.

$2H_2O\ (\ell) \xrightarrow{Electric\ current} 2H_2(g) + O_2\ (g)$

$\underset{\text{Barium chloride}}{BaCl_2(aq)} + H_2SO_4 \longrightarrow \underset{\substack{\text{Barium Sulphate}\\(\text{White Precipitate})}}{BaSO_4} + 2HCl(aq)$

Example: How will you test for the gas which is liberated when hydrochloric acid reacts with an active metal?

Solution: When an active metal like zinc reacts with hydrochloric acid, hydrogen gas is evolved. This is tested by bringing a lighted candle near the gas evolved. It burns with popping sound

$Zn_{(s)} + 2HCl_{(\ell)} \rightarrow ZnCl_{2(aq)} + H_{2(g)} \uparrow$

Zinc — Hydrochloric acid — Zinc chloride

Example: When you mix solutions of lead (II) nitrate and potassium iodide,

(i) What is the colour of the precipitate formed? Name the compound involved.

(ii) Write a balanced chemical equation for the reaction.

(iii) Is this a double displacement reaction?

Solution: (i) The precipitate is yellow in colour. The compound is lead (II) iodide.

(ii) $Pb(NO_3)_{2(aq)} + 2KI_{(aq)} \longrightarrow PbI_{2(s)} + 2KNO_{3(aq)}$

yellowish

(iii) Yes, it is double displacement reaction.

Example: A green salt on heating decomposes to produce a colourless suffocating gas and leaves behind a reddish brown residue. Name the salt and write the decomposition reaction.

Solution: The salt is ferrous sulphate ($FeSO_4.7H_2O$)

$2FeSO_4 \cdot 7H_2O_{(s)} \xrightarrow[-14H_2O]{Heat} 2FeSO_{4(s)} \xrightarrow{Heat} Fe_2O_{3(s)} + SO_{2(g)} + SO_{3(g)}$

Hydrated ferrous sulphate	Anhydrous ferrous sulphate	Ferric oxide (Reddish brown)	Sulphur dioxide	Sulphur trioxide (Colourless Suffocating gas)

CHEMICAL EQUATIONS & BALANCING:

All chemical changes are accompanied by chemical reactions. These reactions can be described in sentence form, but the description would be quite long. Chemical equations have been framed to describe the chemical reactions.

A chemical equation links together the substances which react (reactants) with the new substances that are formed (products).

Zinc + Hydrochloric acid $\longrightarrow$ Zinc chloride + Hydrogen

(Reactants) (Products)

Mind it: A chemical reaction can be summarized by chemical equation.

(a) Types of Chemical Equations:

(i) Word equation: A word equation links together the names of the reactants with those of the products.

For example, the word equation, when magnesium ribbon burns in oxygen to form a white powder of magnesium oxide, may be written as follows-

Magnesium + Oxygen $\longrightarrow$ Magnesium oxide

(Reactants) (Products)

Similarly, the word equation for the chemical reaction between granulated zinc and hydrochloric acid may be written as –

Zinc + Sulphuric acid $\longrightarrow$ Zinc sulphate + Hydrogen

In a word equation

- The reactants are written on the left hand side with a plus sign (+) between them.
- The products are written on the right hand side with a plus sign (+) between them.
- An arrow $\longrightarrow$ separates the reactants from the products.
- The direction of the arrow head points towards the products.

Mind it: Although word equations are quite useful, yet they don't give the true picture of chemical reactions.

(ii) Symbol equation: A brief representation of a chemical reaction in terms of symbols and formulae of the substances involved is known as a symbol equation.

In a symbol equation, the symbols and formulae of the elements and compounds are written instead of their word names.

e.g. Burning of magnesium in oxygen to form magnesium oxide may be written as follows:

$Mg + O_2 \longrightarrow MgO$

Mind it: Symbol equations are always written from the word equations.

(b) Unbalanced and Balanced Chemical Equations:

In an unbalanced equation, the number of atoms of different elements on both sides of the equation are not equal. For example, in the equation given below, the number of Mg atoms on both sides of the equation is one (same), but the number of oxygen atoms are not equal. It is known as an unbalanced equation.

$Mg + O_2 \longrightarrow MgO$

An unbalanced equation is also called skeletal equation.

In a balanced equation, the number of atoms of different elements on both sides of the equation are always equal. The balanced equation for the burning of magnesium ribbon in oxygen is written as –

$2Mg + O_2 \longrightarrow 2MgO$

(i) Importance of balanced chemical equation: The balancing of a chemical equation is essential or necessary to fulfill the requirement of "Law of conservation of mass".

(ii) Balancing of chemical equations: Balancing of chemical equations may be defined as –

"The process of making the number of different types of elements, on both sides of the equation, equal."

The balancing of a chemical equation is done with the help of Hit and Trial method. In this method, the coefficients before the symbols or formulae of the reactants and products are adjusted in such a way that the total number of atoms of each element on both the sides of the arrow head become equal. This balancing is also known as mass balancing because if the atoms of elements on both sides are equal then their masses will also be equal.

The major steps involved in balancing a chemical equation are as follows -

- Write the chemical equation in the form of a word equation. Keep the reactants on the left side and the products on the right side. Separate them by an arrow ($\longrightarrow$), whose head points from the reactants towards the products.
- Convert the word equation into the symbol equation by writing the symbols and formulae of all the reactants and products.
- Make the atoms of different elements on both sides of the equation equal by suitable method. This is known as balancing of equation.
- Do not change the formulae of the substances while balancing the equation.
- Make the equation more informative if possible.

Example: Iron reacts with water (steam) to form iron (II, III) oxide and liberates hydrogen gas.

Solution: The word equation for the reaction is -

Iron + Water $\longrightarrow$ Iron (II, III) oxide + Hydrogen

The symbol equation for the same reaction is -

$Fe + H_2O \longrightarrow Fe_3O_4 + H_2$

The balancing of the equations is done in the following steps:

Example: Zinc reacts with dilute sulphuric acid to give zinc sulphate and hydrogen. Write the chemical equation.

Solution: The word equation for the reaction is -

Zinc + Sulphuric acid $\longrightarrow$ Zinc sulphate + Hydrogen

The symbol equation for the same reaction is -

$Zn + H_2SO_4 \longrightarrow ZnSO_4 + H_2$

Let us count the number of atoms of all the elements in the reactants and products on both sides of the equation.

Element	**No. of atoms of reactants (L.H.S.)**	**No. of atoms of products (R.H.S.)**
Zn	1	1
S	1	1
H	2	2
O	4	4

As the number of atoms of the elements involved in the reactants and products are equal, the equation is already balanced.

I: Let us count the number of atoms of all the elements in the reactants and products on both sides of the equation.

Element	No. of atoms (L.H.S.)	No. of atoms of reactants of products (R.H.S.)
Fe	1	3
H	2	2
O	1	4

Thus, the number of H atoms are equal on both sides. At the same time, the number of Fe and O atoms are not equal.

II: On inspection, the number of O atoms in the reactant (H_2O) is 1 while in the product (Fe_3O_4), these are 4. To balance the atoms, put coefficient 4 before H_2O on the reactant side. The partially balanced equation may be written as

$Fe + 4H_2O \longrightarrow Fe_3O_4 + H_2$

III: In order to equate H atoms, put coefficient 4 before H_2 on the product side. As a result, the H atoms on both sides of the equation become 8 and are thus balanced. The partially balanced equation may now be written as

$Fe + 4H_2O \longrightarrow Fe_3O_4 + 4H_2$

IV: In order to balance the Fe atoms, put coefficient 3 before Fe on the reactant side. The equation formed may be written as -

$3Fe + 4H_2O \longrightarrow Fe_3O_4 + 4H_2$

V: On final inspection, the number of atoms of all the elements on both sides of the equation are equal. Therefore, the equation is balanced.

(c) Specialties of Chemical Equation:

(i) We get the information about the substances which are taking part and formed in the reaction.

(ii) We get the information about the number of molecules of elements or compounds which are either taking part or formed in the chemical reaction.

(iii) We also get the information of weight of reactants or products.

For example - $CaCO_3 \longrightarrow CaO + CO_2$

(100g) (56 g) (44 g)

Total weight of reactants is equal to the total weight of products because matter is never destroyed. In the above example total weight of calcium carbonate (reactant) is 100 g and of product is also 100g (56 g + 44 g).

(iv) In a chemical equation if any reactant or product is in gaseous state, then its volume can also be determined. For example in the above reaction volume of carbon dioxide is 22.4 litres at S.T.P.

(v) In a chemical equation with the help of product we can get information about the valency as well.

For example -

$Mg + 2HCl \longrightarrow MgCl_2 + H_2$

In the above reaction one atom of Mg displaces two atoms of hydrogen, so valency of magnesium is two.

Mind it: All chemical equations are written under N.T.P. conditions (at 298 K and 1 atmosphere pressure) if conditions are not otherwise mentioned.

(d) Limitations of Chemical Equations and their Rectifications :

(i) We do not get information about the physical state of reactants and products.
For example solid, liquid or gas.

(ii) No information about the concentration of reactants and products is obtained .

(iii) No information about the speed of reaction and sense of timing can be obtained.

(iv) Information regarding the favourable conditions of the reactions such as pressure, temperature, catalyst etc. can't be obtained during the reaction.

(v) We do not get information whether energy is absorbed or evolved during the reaction.

(vi) We do not get information whether the reaction is reversible or irreversible.

(vii) We do not get information about the necessary precautions to be taken for the completion of reaction.

The above limitations are rectified in the following manner –

(i) The physical state of reactants and products are represented by writing them in bracket.

Writing State Symbols:

The chemical equations or symbol equations which we have enlisted don't mention the physical states of the reactant and product species involved in the reaction. In order to make the equation more informative, the physical states are also mentioned with the help of certain specific symbols known as state symbols. These symbols are

- (s) for solid state
- (ℓ) for liquid state
- (g) for gaseous state
- (aq) for aqueous solution i.e., solution prepared in water.

Sometimes a gas if evolved in a reaction is shown by the symbol ($\uparrow$) i.e., by an arrow pointing upwards.

Similarly the precipitate, if formed during the reaction, is indicated by the symbol ($\downarrow$) i.e., by an arrow pointing downwards.

Significance of State Symbols:

(i) The state symbols are of most significance for those chemical reactions which are either accompanied by the evolution of heat (exothermic) or by the absorption of heat (endothermic). For example.

$2H_2(g) + O_2(g) \longrightarrow 2H_2O(\ell) + 572$ kJ

$2H_2(g) + O_2(g) \longrightarrow 2H_2O(g) + 484$ kJ

Both these reactions are of exothermic nature because heat has been evolved in these. However, actual amounts of heat evolved are different when water is in the liquid state and when it is in the vapour state.

(ii) The precipitate formed in the reaction is represented by ↓ symbol and gaseous substance by ↑ symbol.

The abbreviation 'ppt' is also used to represent the precipitate, if formed.

$2Na(s) + 2H_2O(\ell) \longrightarrow 2NaOH(aq) + H_2(g)$ or $H_2 \uparrow$

$Ca(OH)_2(aq) + CO_2(g) \longrightarrow CaCO_3 \downarrow + H_2O(\ell)$

$AgNO_3(aq) + NaCl(aq) \longrightarrow AgCl$ (ppt.) $+ NaNO_3(aq)$

(iii) To express the concentration, dilute or conc. is written below the formula.

$Mg + H_2SO_4 \longrightarrow MgSO_4 + H_2$

(dilute)

(iv) Favourable conditions required for the completion of reaction are written above and below the arrow.

$N_2 + 3H_2 \xrightarrow[500°C, 200\ atm]{Fe/M_0} 2NH_3 + 22400$ Cal. heat

(v) Reversible reaction is represented by ( ) symbol and irreversible reaction by (⟶) symbol.

(vi) The heat absorbed in the chemical reaction is written on the right side by putting negative (–) sign and heat evolved in the chemical reaction is written on the right side by putting positive (+)sign.

$N_2 + 3H_2 \longrightarrow 2NH_3 + 22400$ Calorie

(Exothermic Reaction)

$N_2 + O_2 \longrightarrow 2NO - 43200$ Calorie

(Endothermic Reaction)

Example: A housewife wanted her house to be white washed. She bought 10 kg of quick lime from the market and dissolved it in 30 litres of water. On adding quick lime to water, she noticed that the water started boiling even when it was not being heated. Give reason for the observation. Write the corresponding chemical equation and name the product formed.

Solution: Water started boiling because the reaction of quick lime with water is highly exothermic.

$CaO(s) + H_2O(\ell) \longrightarrow Ca(OH)_2(aq) + Heat$

Quick Lime — Water — Calcium hydroxide (Slaked lime)

Thus, the product formed is slaked lime (calcium hydroxide).

Example: Write balanced equations that predict the reactions of cesium metal with

(a) $Cl_2(g)$ (b) $H_2O(\ell)$ (c) $H_2(g)$

Solution: Cesium is an alkali metal. We therefore expect that its chemistry will be dominated by oxidation of the metal to Cs^+ ions. Further, we recognize that Cs is far down the periodic table, which means it will be among the most active of all metals and will probably react with all three of the substances listed.

The reaction between Cs and Cl_2 is a simple combination reaction two elements, one a metal and one a non-metal, forming the ionic compound CsCl:

$2Cs(s) + Cl_2(g) \longrightarrow 2CsCl\ (s)$.

We predict the reactions of cesium with water and hydrogen to proceed as follows:

$2Cs(s) + 2H_2O(\ell) \longrightarrow 2CsOH(aq) + H_2(g)$

$2Cs(s) + H_2(g) \longrightarrow 2CsH(s)$

In each case cesium forms a Cs^+ ion in its compound. The chlorine (Cl^-), hydroxide (OH^-), and hydride (H^-) ions are all univalent, so the final products have 1 : 1 stoichiometry with Cs^+.

Example: Represent each of the following word equations with a balanced chemical equation.

(a) Disilane gas (Si_2H_6) undergoes combustion to form solid silicon dioxide and water.

(b) Solid aluminium hydride is formed by a combination reaction of its two elements.

(c) When solid calcium bisulphite is heated, it decomposes to solid calcium oxide, sulphur dioxide gas, and water.

Solution: (a) $2Si_2H_6\ (g) + 7O_2(g) \longrightarrow 4SiO_2(s) + 6H_2O(\ell)$

(b) $2Al(s) + 3H_2(g) \longrightarrow 2AlH_3(s)$

(c) $Ca(HSO_3)_2\ (s) \xrightarrow{\Delta} CaO(s) + 2SO_2(g) + H_2O(\ell)$.

OXIDATION & REDUCTION

Let us do a comparative study of oxidation and reduction:

Oxidation	Reduction
1. Addition of Oxygen	1. Removal of Oxygen
e.g. $2Mg + O_2 \rightarrow 2MgO$	e.g. $CuO + C \rightarrow Cu + CO$
2. Removal of Hydrogen	2. Addition of Hydrogen
e.g. $H_2S + Cl_2 \rightarrow 2HCl + S$	e.g. $S + H_2 \rightarrow H_2S$
3. Increase in positive charge	3. Decrease in positive charge
e.g. $Fe^{2+} \rightarrow Fe^{3+} + e^-$	e.g. $Fe^{3+} + e^- \rightarrow Fe^{2+}$
4. Increase in oxidation number	4. Decrease in oxidation number
(+2) (+4)	(+7) (+2)
e.g. $SnCl_2 \rightarrow SnCl_4$	e.g. $MnO_4^- \rightarrow Mn^{2+}$
5. Removal of electron	5. Addition of electron
e.g. $Sn^{2+} \rightarrow Sn^{4+} + 2e^-$	e.g. $Fe^{3+} + e^- \rightarrow Fe^{2+}$

(a) Oxidation Number

- It is an imaginary or apparent charge developed over atom of an element when it goes from its elemental free state to combined state in molecules.
- It is calculated on basis of an arbitrary set of rules.
- It is a relative charge in a particular bonded state.
- In order to keep track of electron-shifts in chemical reactions involving formation of compounds, a more practical method of using oxidation number has been developed.
- In this method, it is always assumed that there is a complete transfer of electron from a less electronegative atom to a more electronegative atom.

Rules governing oxidation number

The following rules are helpful in calculating oxidation number of the elements in their different compounds. It is to be remembered that the basis of these rule is the electro negativity of the element.

- **Fluorine atom:**
  Fluorine is most electronegative atom (known). It always has oxidation number equal to –1 in all its compounds

- **Oxygen atom:**
  In general and as well as in its oxides, oxygen atom has oxidation number equal to –2.
  In case of
  (i) peroxide (e.g. H_2O_2, , Na_2O_2) is –1,
  (ii) super oxide (e.g. KO_2) is –1/2
  (iii) ozonide (e.g. KO_3) is –1/3
  (iv) in OF_2 is + 2 & in O_2F_2 is +1

- **Hydrogen atom:**
  In general, H atom has oxidation number equal to +1. But in metallic hydrides (e.g. NaH, KH), it is –1.

- **Halogen atom:**
  In general, all halogen atoms (Cl, Br, I) have oxidation number equal to –1.
  But if halogen atom is attached with a more electronegative atom than halogen atom, then it will show positive oxidation numbers.
  e.g. $K\overset{+5}{Cl}O_3$, $H\overset{+5}{I}O_3$, $H\overset{+7}{Cl}O_4$, $K\overset{+5}{Br}O_3$

- **Metals:**
  (a) Alkali metal (Li, Na, K, Rb, ...) always have oxidation number +1
  (b) Alkaline earth metal (Be, Mg, Ca .....) always have oxidation number +2.
  (c) Aluminium always has +3 oxidation number

Mind it: Metal may have negative or zero oxidation number

- Oxidation number of an element in free state or in allotropic forms is always zero
  e.g. $\overset{0}{O_2}$, $\overset{0}{S_8}$, $\overset{0}{P_4}$, $\overset{0}{O_3}$
- Sum of the oxidation numbers of atoms of all elements in a molecule is zero.
- Sum of the oxidation numbers of atoms of all elements in an ion is equal to the charge on the ion.
- If the group number of an element in modern periodic table is n, then its oxidation number may vary from (n – 10) to (n – 18) (but it is mainly applicable for p-block elements)
  e.g. N- atom belongs to 15th group in the periodic table, therefore as per rule, its oxidation number may vary from –3 to +5 ($\overset{-3}{N}H_3$, $\overset{-2}{N}O$, $\overset{+3}{N}_2O_3$, $\overset{-4}{N}O_2$, $\overset{-5}{N}_2O_2$)
- The maximum possible oxidation number of any element in a compound is never more than the number of electrons in valence shell.(but it is mainly applicable for p-block elements)

Calculation of average oxidation number:

Example: Calculate oxidation number of underlined element:

(a) $Na_2 \underline{S}_2O_3$ (b) $Na_2 \underline{S}_4O_6$

Solution: **(a)** Let oxidation number of S-atom is x. Now work accordingly with the rules given before.

$\therefore$ $(+1) \times 2 + (x) \times 2 + (-2) \times 3 = 0$

$x = +2$

(b) Let oxidation number of S-atom is x

$\therefore$ $(+1) \times 2 + (x) \times 4 + (-2) \times 6 = 0$

$x = +2.5$

- It is important to note here that $Na_2S_2O_3$ have two S-atoms and there are four S-atom in $Na_2S_4O_6$. However none of the sulphur atoms in both the compounds have + 2 or + 2.5 oxidation number, it is the average of oxidation number, which reside on each sulphur atom. Therefore, we should work to calculate the individual oxidation number of each sulphur atom in these compounds.

Calculation of individual oxidation number

It is important to note that to calculate individual oxidation number of the element in its compound one should know the structure of the compound and use the following guidelines.

Formula:

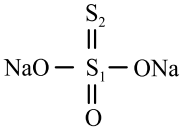
Oxidation Number = Number of electrons in the valence shell – Number of electrons taken up after bonding

Guidelines: It is based on electronegativity of elements.

1. If there is a bond between similar type of atom and each atom has same type of hybridization, then bonded pair electrons are equally shared by each element.

Example: Find the oxidation state of S atoms present in $Na_2S_2O_3$.

Solution: Some times same atom in a compound has different oxidation state for example, structures of $Na_2S_2O_3$ is



Here S_1 and S_2 both are sulphur atoms but they have different oxidation state

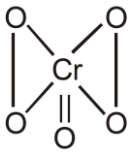
Oxidation state of $S_1 = 4$

Oxidation state of $S_2 = 0$

Average oxidation state of S = $\frac{4+0}{2} = 2$

Miscellaneous Examples:

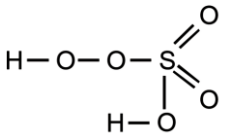
In order to determine the exact or individual oxidation number we need to take help from the structures of the molecules. Some special cases are discussed as follows:

- The structure of CrO_5 is 

From the structure, it is evident that in CrO_5 there are two peroxide linkages and one double bond. The contribution of each peroxide linkage is –2. Let the oxidation number of Cr is x.

$\therefore$ $x + (-2)2 + (-2) = 0$ or $x = 6$

$\therefore$ Oxidation number of Cr = + 6 **Ans.**

- The structure of H_2SO_5 is 

From the structure, it is evident that in H_2SO_5, there is one peroxide linkage, two sulphur-oxygen double bonds and one OH group. Let the oxidation number of S = x.

$\therefore$ $(+1) + (-2) + x + (-2)\,2 + (-2) + 1 = 0$

or $x + 2 - 8 = 0$ or $x - 6 = 0$ or $x = 6$

$\therefore$ Oxidation number of S in H_2SO_5 is + 6 **Ans.**

(b) Oxidising and reducing agent

(i) Oxidising agent or Oxidant:

Oxidising agents are those compounds which can oxidise others and reduce itself during the chemical reaction. Those reagents in which for an element, oxidation number decreases or which undergoes gain of electrons in a redox reaction are termed as oxidants.

General Characterstics of Oxidising agent

(a) Oxidising agents are lewis acids (tendency to accept electron pair).

(b) Substances which can oxidise others and get reduce themselves.

(c) Substances which show the decrement in oxidation number.

Some important oxidising agents or oxidants

(i) All high electronegative elements like N, O, F, Cl etc.

(ii) All metallic oxides like Li_2O, Na_2O, Na_2O_2, CO_2, CaO, MgO, BaO_2 etc.

(iii) Some nonmetallic oxides like CO_2, SO_2, H_2O_2, O_3.

(iv) All neutral compound or ion in which element shows their highest oxidation no or state, will act as oxidant or oxidising agent like $KMnO_4$, H_2SO_4, $SnCl_4$, H_3PO_4, $K_2Cr_2O_7$, $HClO_4$, $CuCl_2$, HNO_3, H_2SO_5, $FeCl_3$, $HgCl_2$, etc.

(v) $KMnO_4$, $K_2Cr_2O_7$, HNO_3, conc.H_2SO_4 etc are powerful oxidising agents.

(ii) Reducing agent or Reductant:

Reducing agents are those compounds which can reduce other and oxidise itself during the chemical reaction. Those reagents in which for an element, oxidation number increases or which undergoes loss of electrons in a redox reaction are termed as reductants.

General Characteristics of Oxidising agent

(a) Reducing agents are lewis bases (tendency to lose electron pair).

(b) Substances which can reduce other and get oxidise themselves.

(c) Substances which show the increment in oxidation number.

Some important reducing agents or reductants

(a) All metals like, K, Mg, Ca, etc.

(b) All metallic hydrides like NaH, CaH_2, $LiAlH_4$, $NaBH_4$, AlH_3 etc.

(c) All hydroacids like HF, HCl, HBr, H_2S, etc.

(d) Some organic compounds like Aldehydes, formic acid, oxalic acid, tartaric acid.

(e) All neutral compounds or ions, in which element shows their lowest oxidation no. or state, will ad as reductant or reducing agent MnO, HClO, $HClO_2$, $H3PO_2$, HNO_2, H_2SO_3, $FeCl_2$, $SnCl_2$, Hg_2Cl_2, CH_2Cl_2 etc.

Some important compound which can acts as oxidant and reductant both: HNO_2, SO_2, H_2O_2, O_3, Al_2O_3, CrO_2, MnO_2, ZnO, CuO

Mind it: Al_2O_3, CrO_2, MnO_2, ZnO, CuO are called as amphoteric oxide.

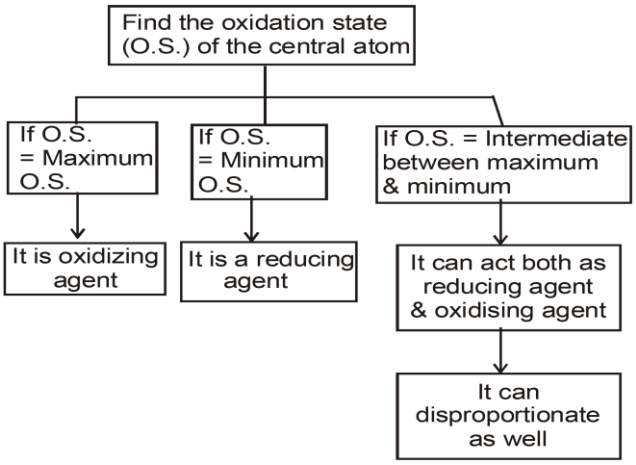
e.g. KI, $Na_2S_2O_3$ etc. are the powerful reducing agents.

Note: There are some compounds also which can work both as oxidising agent and reducing agent

e.g. H_2O_2, NO_2

- Oxidising strength $\propto$ Oxidation Number
- Reducing strength $\propto \frac{1}{\text{OxidationNumber}}$

HOW TO IDENTIFY WHETHER A PARTICULAR SUBSTANCE IS AN OXIDISING OR A REDUCING AGENT



(c) Application of Oxidation Number

(1) To compare the strength of oxyacid and base:-

Strength of acid α Oxidation Number of central element

Strength of base α $\frac{1}{\text{Oxidation Number of central element}}$

Example: Order of acidic strength in $HClO$, $HClO_2$, $HClO_3$, $HClO_4$ will be.

Solution:

Oxidation	Number of chlorine
$HClO$ (Hypo chlorus acid)	+1
$HClO_2$ (Chlorous acid)	+3
$HClO_3$ (Chloric acid)	+5
$HClO_4$ (Perchloric acid)	+7

Strength of acid α Oxidation Number

So the order will be $HClO_4 > HClO_3 > HClO_2 > HClO$

(2) To determine the oxidising and reducing nature:

Group	**Range of Oxidation Number**(n – 8) to n (n = valence electron)
I A	+1
II A	+2
III A	+1, +3
IVA	–4 to+4
VA	–3 to +5
VI A	–2 to +6[Exception= Maximum Oxidation Number of Oxygen is 2]
VII A	–1 to +7 [Exception = Oxidation of F is –1 in combined state]

Example: Identify the correct statement (s) in relation to the following reaction:

$Zn + 2HCl \rightarrow ZnCl_2 + H_2$

(A) Zn is acting as an oxidant (B) Chlorine is acting as a reductant

(C) Hydrogen ion is acting as an oxidant (D) Zinc is acting as a reductant

Solution: (D) In the reaction Zn is converted in to Zn^{+2} by the loss of electrons & itself get oxidized.

Example: Which substance is reduced in the following reaction?

$ZnO + C \rightarrow Zn + CO$

(A) Carbon (B) Zinc oxide (C) Zinc (D) Carbon monoxide

Solution: (B) ZnO loses its oxygen and converted in to Zn metal. loss of oxygen from a molecule is called reduction. so ZnO is reduced in this reaction & carbon get oxidized.

Example: An oxidizing agent is a substance which: a. loses oxygen b. gains hydrogen c. gains oxygen

(A) only a (B) only b (C) only c (D) Both a & b

Solution: (D) Oxidizing agent is a substance which reduced itself. It loses oxygen, gains hydrogen or loses electrons.

REDOX REACTIONS

A reaction in which oxidation and reduction simultaneously take place is called a redox reaction in all redox reactions, the total increase in oxidation number must be equal to the total decrease in oxidation number.

e.g. $10\overset{+2}{Fe}SO_4 + 2K\overset{+7}{Mn}O_4 + 8H_2SO_4 \longrightarrow 5\overset{+2}{Fe}_2(SO_4)_3 + 2\overset{+2}{Mn}SO_4 + K_2SO_4 + 8H_2O$

(a) Disproportionation reaction

A redox reaction in which same element present in a particular compound in a definite oxidation state is oxidized as well as reduced simultaneously is a disproportionation reaction.

Disproportionation reactions are a special type of redox reactions. One of the reactants in a disproportionation reaction always contains **an element that can exist in at least three oxidation states.** The element in the form of reacting substance is in the intermediate oxidation state and both higher and lower oxidation states of that element are formed in the reaction. For example:

$$2\overset{-1}{H_2O_2}(aq) \longrightarrow 2\overset{+2}{H_2O}(\ell) + \overset{0}{O_2}(g)$$

$$\overset{0}{S_8}(s) + 12OH^-(aq) \longrightarrow 4\overset{-2}{S^{2-}}(aq) + 2\overset{+2}{S_2}O_3^{2-}(aq) + 6H_2O(\ell)$$

$$\overset{0}{Cl_2}(g) + 2OH^-(aq) \longrightarrow 4\overset{+1}{Cl}O^-(aq) + H_2O(\ell)$$

Consider the following reactions:

(a) $2KClO_3 \longrightarrow 2KCl + 3O_2$

$KClO_3$ plays a role of oxidant and reductant both. Here, Cl present in $KClO_3$ is reduced and O present in $KClO_3$ is oxidized. Since same element is not oxidized and reduced, **so it is not a disproportionation reaction, although it looks like one.**

(b) $NH_4NO_2 \longrightarrow N_2 + 2H_2O$

Nitrogen in this compound has –3 and +3 oxidation number, which is not a definite value. So, it is not a disproportionation reaction. It is an example of **comproportionation reaction**, which is a class of redox reaction in which an element from two different oxidation state gets converted into a single oxidation state.

(c) $4K\overset{+5}{Cl}O_3 \longrightarrow 3K\overset{+7}{Cl}O_4 + K\overset{-1}{Cl}$

It is a case of disproportionation reaction and Cl atom is disproportionating.

List of some important disproportionation reactions

1. $H_2O_2 \longrightarrow H_2O + O_2$
2. $X_2 + OH^-$ (dil.) $\longrightarrow X^- + XO^-$ (X = Cl, Br, I)
3. $X_2 + OH^-$ (conc.) $\longrightarrow X^- + XO_3^-$

   F_2 does not undergo disproportionation as it is the most electronegative element.

   $F_2 + NaOH(dil.) \longrightarrow F^- + OF_2$

   $F_2 + NaOH(conc.) \longrightarrow F^- + O_2$
4. $(CN)_2 + OH^- \longrightarrow CN^- + OCN^-$
5. $P_4 + OH^- \longrightarrow PH_3 + H_2PO_2^-$
6. $S_8 + OH^- \longrightarrow S^{2-} + S_2O_3^{2-}$
7. $MnO_4^{2-} \longrightarrow MnO_4^- + MnO_2$
8. $NH_2OH \longrightarrow N_2O + NH_3$

   $NH_2OH \longrightarrow N_2 + NH_3$
9. Oxyacids of Phosphorus (+1, +3 oxidation number)

   $H_3PO_2 \longrightarrow PH_3 + H_3PO_3$

   $H_3PO_3 \longrightarrow PH_3 + H_3PO_4$

10. Oxyacids of Chlorine (Halogens) (+1, +3, +5 Oxidation number)

$ClO^- \longrightarrow Cl^- + ClO_2^-$

$ClO_2^- \longrightarrow Cl^- + ClO_3^-$

$ClO_3^- \longrightarrow Cl^- + ClO_4^-$

11. $HNO_2 \longrightarrow NO + HNO_3$

Comproportionation reaction:

Reverse of disproportionation is called Comproportionation. In some of the disproportionation reactions, by changing the medium (from acidic to basic or reverse), the reaction goes in backward direction and can be taken as an example of Comproportionation reaction.

$I^- + IO_3^- + H^+ \longrightarrow I_2 + H_2O$

(b) Balancing of redox reactions

All balanced equations must satisfy two criteria.

1. Atom balance (mass balance):

There should be the same number of atoms of each kind on reactant and product side.

2. Charge balance:

The sum of actual charges on both sides of the equation must be equal.

There are two methods for balancing the redox equations:

1. Oxidation - number change method
2. Ion electron method or half cell method

- Since First method is not very much fruitful for the balancing of redox reactions, students are advised to use second method (Ion electron method) to balance the redox reactions.

Ion electron method:

By this method redox equations are balanced in two different medium.

(a) Acidic medium (b) Basic medium

- **Balancing in acidic medium**

Students are advised to follow the following steps to balance the redox reactions by Ion electron method in acidic medium.

Example: Balance the following redox reaction:

$FeSO_4 + KMnO_4 + H_2SO_4 \rightarrow Fe_2(SO_4)_3 + MnSO_4 + H_2O + K_2SO_4$

Solution:

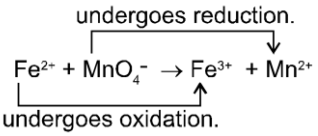
Step – I: Assign the oxidation number to each element present in the reaction.

$$\overset{+2\ +6\ -2}{Fe\,S\,O_4} + \overset{+1\ +7\ -2}{K\,Mn\,O_4} + \overset{+1\ +6\ -2}{H_2\,S\,O_4} \longrightarrow \overset{+3\ +6\ -2}{Fe_2(S\,O_4)_3} + \overset{+1\ -2}{H_2\,O} + \overset{+1\ +6\ -2}{K_2\,S\,O_4} + \overset{+2\ +6\ -2}{Mn\,S\,O_4}$$

Step – II: Now convert the reaction in ionic form by eliminating the elements or species, which are not undergoing either oxidation or reduction.

$$Fe^{2+} + \overset{+7}{Mn}\,O_4^- \longrightarrow Fe^{3+} + Mn^{2+}$$

Step – III: Now identify the oxidation / reduction occurring in the reaction



Step – IV: Split the Ionic reaction in two half, one for oxidation and other for reduction.

$$Fe^{2+} \xrightarrow{\text{Oxidation}} Fe^{3+} \;\Big|\; MnO_4^- \xrightarrow{\text{Reduction}} Mn^{2+}$$

Step – V: Balance the atom other than oxygen and hydrogen atom in both half reactions

$$Fe^{2+} \longrightarrow Fe^{3+} \;\Big|\; MnO_4^- \longrightarrow Mn^{2+}$$

Fe & Mn atoms are balanced on both side.

Step – VI: Now balance O & H atom by H_2O & H^+ respectively by the following way: For one excess oxygen atom, add one H_2O on the other side and two H^+ on the same side.

$Fe^{2+} \longrightarrow Fe^{3+}$ (no oxygen atom) .....(i)

$8H^+ + MnO_4^- \longrightarrow Mn^{2+} + 4H_2O$ ....(ii)

Step – VII: Equation (i) & (ii) are balanced atom wise. Now balance both equations charge wise. To balance the charge, add electrons to the electrically positive side.

$Fe^{2+} \xrightarrow{\text{oxidation}} Fe^{3+} + e^-$ .....(1)

$5e^- + 8H^+ + MnO_4 \xrightarrow{\text{Reduction}} Mn^{2+} + 4H_2O$ .....(2)

Step – VIII: The number of electrons gained and lost in each half - reaction are equalized by multiplying both the half reactions with a suitable factor and finally the half reactions are added to give the overall balanced reaction.

Here, we multiply equation (1) by 5 and (2) by 1 and add them:

$Fe^{2+} \longrightarrow Fe^{3+} + e^-$ ..... (1) × 5

$$\frac{5e^- + 8H^+ + MnO_4^- \longrightarrow Mn^{2+} + 4H_2O \ldots\ldots\ldots (2) \times 1}{5Fe^{2+} + 8H^+ + MnO_4^- \longrightarrow 5Fe^{3+} + Mn^{2+} + 4H_2O}$$

(Here, at this stage, you will get balanced redox reaction in Ionic form)

Step – IX: Now convert the Ionic reaction into molecular form by adding the elements or species, which are removed in step (2).

Now, by some manipulation, you will get:

$$5\ FeSO_4 + KMnO_4 + 4H_2SO_4 \longrightarrow \frac{5}{2}\ Fe_2(SO_4)_3 + MnSO_4 + 4H_2O + \frac{1}{2}\ K_2SO_4$$

or $10FeSO_4 + 2KMnO_4 + 8H_2SO_4 \longrightarrow 5Fe_2(SO_4)_3 + 2MnSO_4 + 8H_2O + K_2SO_4$.

- **Balancing in basic medium:**

In this case, except step VI, all the steps are same. We can understand it by the following example:

Example: Balance the following redox reaction in basic medium:

$ClO^- + CrO_2^- + OH^- \longrightarrow Cl^- + CrO_4^{2-} + H_2O$

Solution: By using upto step V, we will get:

$$\overset{+1}{Cr}O^- \xrightarrow{\text{Reduction}} Cl^- \;\Big|\; \overset{+3}{Cr}O_2^- \xrightarrow{\text{Oxidation}} \overset{+6}{Cr}O_4^{2-}$$

Now, students are advised to follow step VI to balance 'O' and 'H' atom.

$2H^+ + ClO^- \rightarrow Cl^- + H_2O \mid 2H_2O + CrO_2^- + CrO_4^{2-} + 4H^+$

- Now, since we are balancing in basic medium, therefore add as many as OH^- on both side of equation as there are H+ ions in the equation.

  $2OH^- + 2H^+ + ClO^- \longrightarrow Cl^- + H_2O + 2OH^-$

  $4OH^- + 2H_2O + CrO_2^- \longrightarrow CrO_4^{2-} + 4H^+ + 4OH^-$

  Finally, you will get Finally you will get

  $H_2O + ClO^- \longrightarrow Cl^- + 2OH^-$ .....(i) $4OH^- + CrO_2^- \longrightarrow CrO_4^{2-} + 2H_2O$ ..... (ii)

  Now see equation (i) and (ii) in which O and H atoms are balanced by OH^- and H_2O

  Now from step VIII

  $2e^- + H_2O + ClO^- \longrightarrow Cl^- + 2OH^-$ ..... (i) ×3

  $4OH^- + CrO_2^- \longrightarrow CrO_4^{2-} + 2H_2O + 3e^-$.... (ii) ×2

  Adding: $3ClO^- + 2CrO_2^- + 2OH^- \longrightarrow 3Cl^- + 2CrO_4^{2-} + H_2O$

APPLICATION OF REDOX REACTION:

Redox reactions are the basis of many applications in chemistry. Some of the important applications of redox reactions are:

- **Photosynthesis**

  Photosynthesis is a process which to converts carbon dioxide and water by chlorophyll of green plants in the presence of sunlight to carbohydrates.

  $6CO_2(g) + 6H_2O(\ell) \xrightarrow[\text{Sunlight}]{\text{Chlorophyl}} C_6H_{12}O_{6(aq)} + 6O_2(g)$

  In this case, CO_2 is reduced to carbohydrates and water is oxidized to oxygen. The light provides the energy required for the reaction. This reaction is a source of food for plants and animals.

- **In metallurgy**

  Many metal oxides are reduced to metals by using suitable reducing agents. For example, Al_2O_3 is reduced to aluminium by cathodic reduction in electrolytic cell. Fe_2O_3 is reduced to iron in a blast furnace using coke. Other metals such as magnesium, calcium, strontium, barium, sodium, potassium, etc. are produced commercially by electrolysis methods.

- **Production of Chemicals**

  Many chemical compounds of our common need are commercially prepared by redox reactions. For example, sodium, hydroxide, Cl_2, F_2, KOH, etc. are produced by electrolysis which is based on redox reactions.

- **Oxidation of Fuels**

  Oxidation of fuels is an important source of energy which satisfies our daily need of life. These are used for obtaining energy for domestic, transport or industrial purposes. They are also used for digestion of food in animals.

  Fuels + $O_2 \rightarrow CO_2 + H_2O$ + Energy

  In living cells, glucose ($C_6H_{12}O_6$) is oxidized to CO_2 and H_2O in the presence of oxygen and energy is released

  $C_6H_{12}O_{6(aq)} + 6O_{2(g)} \rightarrow 6CO_{2(g)} + 6H_2O_{(\ell)}$ + Energy

- **Electrochemical Cells or Batteries**

  The electrochemical cells or batteries are most important and are contributing significantly in different fields. They provided provide a source of electricity in many motors in small and large equipment's. Our vehicles cars, trucks, buses, etc. are moving on the storage cells involving reaction between hydrogen and oxygen using hydrogen and oxygen electrodes in fuel cells. They meet our demand of electrical energy in space capsule.

- **Quantitative Relationships and Stoichiometry**

  Redox reactions also play significant role in quantitative analysis of different species by redox titrations. These titrations involve the reactions between oxidizing and reducing agents and help to calculate the amount of unknown species in solutions.

Example: Why does blue colour of copper sulphate solution start fading when a zinc rod is dipped in it?

Solution: Zinc is more reactive that copper. It displaces copper from copper sulphate solution to form zinc sulphate, which is colourless. Hence, the blue colour of copper sulphate solution start fading.

$Zn_{(s)}$	+	$CuSO_{4(aq)}$	$\longrightarrow$	$ZnSO_{4(aq)}$	+	Cu
Zinc		Copper sulphate (blue)		Zinc sulphate (colourless)		Copper

Example: A silver spoon is kept immersed in an aqueous copper sulphate solution. What change will take place?

Solution : No change will take place and the silver spoon will remain unaffected. Moreover, blue colour of copper sulphate will also not fade. This is because of the reason that no chemical reaction takes place between silver and copper sulphate solution as silver is placed below copper in the activity series.

Example: With the help of an activity show that iron is more reactive than copper.

Solution: 1. In a test tube, take an aqueous solution of copper sulphate. It is blue in colour.

2. Dip two nails of iron in the solution. Tie the nails to the iron stand with the help of threads.
3. Allow the tube to remain undisturbed for about half an hour.
4. On examination, the blue colour of the solution is found to change into light green. At the same time, a brown deposit appears on the nails. Explanation: Iron is placed above copper in the reactivity series. Therefore, it has displaced copper from copper sulphate solution. The brown deposit on the iron nails is of copper.

Fe(s)	+	$CuSO_4(aq)$	$\longrightarrow$	$FeSO_4(aq)$	+	Cu(s)
(Iron nail)		(Blue)		(Light green)		(Brown)

EFFECTS OF OXIDATION IN OUR DAILY LIFE:

We are all aware of the fact that oxygen is most essential for sustaining life. One can live without food or even water for a number of days but not without oxygen. It is involved in a variety of actions which have wide range of effects on our daily life. Most of them are quite useful while a few may be harmful in nature. Some of these effects are briefly discussed.

(a) Combustion Reactions:

A chemical reaction in which a substance burns or gets oxidized in the presence of air or oxygen is called combustion reaction. For example, kerosene, coal, charcoal, wood etc. burn in air and thus, undergo combustion. Methane (CH_4) a major constituent of natural gas undergoes combustion in excess of oxygen upon heating.

$CH_{4(g)} + 2O_{2(g)} \rightarrow CO_{2(g)} + 2H_2O_{(\ell)}$ + energy

Methane

Similarly, butane (C_4H_{10}), the main constituent of L.P.G. also undergoes combustion.

$2C_4H_{10(g)} + 13O_{2(g)} \rightarrow 8CO_{2(g)} + 5H_2O_{(g)}$ + energy

Butane

All combustion reactions are of exothermic nature and are accompanied by release of heat energy. The human body may be regarded as a furnace or machine in which various food stuffs that we eat undergo oxidation. The heat energy evolved keeps our body working. Carbohydrates such as glucose, fructose, starch etc. are the major source of energy to the human body. They undergo oxidation with the help of oxygen that we inhale to form carbon dioxide and water. For example.

$C_6H_{12}O_{6(s)} + 6O_{2(g)} \rightarrow 6CO_{2(g)} + 6H_2O_{(\ell)}$ + energy

All combustion reactions are not accompanied by flame. Combustion is basically oxidation accompanied by release of energy.

(b) Respiration:

Respiration is the most important biochemical reaction which releases energy in the cells. When we breathe in air, oxygen enters our lungs and passes into thousands of small air sacs (alveoli). These air sacs occupy a large area of membranes and oxygen diffuses from the membranes into blood. It binds itself to haemoglobin present in red blood cells and is carried to millions of cells in the body. Respiration occurs in these cells and is accompanied by the combustion of glucose producing carbon dioxide and water. Since the reaction is of exothermic nature, the energy released during respiration carry out many cell reactions and also keeps our heart and muscles working. It also provides the desired warmth to the body. Both carbon dioxide and water pass back into the blood and we ultimately breathe them out. Respiration takes place in the cells of all living beings.

Mind it: Fish take up oxygen dissolved in water through their gills while plants take up air through small pores (stomata) present in their leaves.

(c) Harmful Effects of Oxidation:

We have discussed the utility of oxidation in releasing energy which our body needs to keep warm and working. However, oxidation has harmful effects also.

(i) Pollution:

The environmental pollution is basically due to combustion. Poisonous gases like carbon monoxide (CO), sulphur dioxide (SO_2), sulphur trioxide (SO_3) and oxides of nitrogen (N_xO_y) etc. are being released into the atmosphere as a result of variety of combustion reaction which are taking place. They pollute the atmosphere and make our lives miserable.

(ii) Corrosion:

The process of slowly eating up of the metals due to attack of atmospheric gases such as oxygen, carbon dioxide, hydrogen sulphide, water vapour etc. on the surface of the metals so as to convert the metal into oxide, carbonate, sulphide etc. is known as corrosion.

The most common example of corrosion is rusting i.e. corrosion of iron. When an iron article remains exposed to moist air for a long time, its surface is covered with a brown, flaky (non-sticky) substance called rust. Rust is mainly hydrated ferric oxide ($Fe_2O_3.xH_2O$). It is formed due to attack of oxygen gas and water vapour present in the air on the surface of iron.

$2Fe(s) + 3/2O_2(g) + xH_2O(l) \rightarrow$ $Fe_2O_3.xH_2O(s)$

hydrated ferric oxide

(Rust)

Similarly, copper objects lose their lustre or shine after some time. The surface of these objects acquire a green coating of basic carbonate, when exposed to air. This is due to attack of O_2, CO_2 and water vapour present in the air on the surface of copper.

$2Cu(s) + CO_2(g) + O_2(g) + H_2O(l) \longrightarrow CuCO_3.Cu(OH)_2$

Copper — Basic copper carbonate (Green)

Likewise, the surface of silver metal gets tarnished (i.e. loses lustre and becomes dull) on exposure to air.

This is due to the formation of a coating of black silver sulphide (Ag_2S) on its surface by the action of H_2S present in the air.

Rusting is a serious problem because it weakens the structure of bridges, iron railings, automobile parts etc. Every year a large amount of money is spent to replace rusted iron and steel structures. The reason is that the reddish brown crust of rust does not stick to the surface. It peels off (or falls down) exposing fresh surface for rusting. Thus, corrosion of iron is a continuous process which ultimately eats up the whole iron object.

Methods to prevent rusting. Rusting can be prevented if iron objects are not allowed to come in contact with the damp air. Some common methods generally used are given below.

(i) By painting the iron articles such as window grills, iron gates, steel furniture, railway coaches, bodies of cars, buses etc.

(ii) By greasing and oiling the iron articles such as mechanical tools, machine parts etc.

(iii) By galvanization, i.e. coating the surface of iron objects with a thin layer of zinc.

(iii) Rancidity:

Oxidation also has damaging effect on foods containing fats and oils. When the food materials prepared in fats and oils are kept for a long time, they start giving unpleasant smell and taste, this is called rancidity.

This happens as follows:

When the fats and oils present in food materials get oxidized by the oxygen (of air), their oxidation products have unpleasant smell and taste. The condition produced by aerial oxidation of fats and oils in foods marked by unpleasant smell and taste is called rancidity. Rancidity spoils the food materials prepared in fats and oils which have been kept for a considerable time and makes them unfit for eating. Rancidity is called 'vikritgandhita' in Hindi.

The development of rancidity of food can be prevented or retarded (slowed down) in the following ways:

1. Rancidity can be prevented by adding anti-oxidants to foods containing fats and oils:

Anti-oxidant is a substance (or chemical) which prevents oxidation. Anti-oxidants are actually reducing agents. When anti-oxidants are added to foods, then the fats and oils present in them do not get oxidized easily and hence do not turn rancid. So, the foods remain good to eat for a much longer time. The two common anti-oxidants used in foods to prevent the development of rancidity are BHA (Butylated Hydroxy- Anisole) and BHT (Butylated Hydroxy Toluene).

2. Rancidity can be prevented by packaging fat and oil containing foods in nitrogen gas:

When the packed food is surrounded by an unreactive gas like nitrogen, there is no oxygen (of air) to cause its oxidation and make it rancid. The manufacturers of potato chips (and other similar food products) fill the plastic bags containing chips with nitrogen gas to prevent the chips from being oxidized and turn rancid.

3. Rancidity can be retarded by keeping food in a refrigerator:

The refrigerator has a low temperature inside it. When the food is kept in a refrigerator, the oxidation of fats and oils in it is slowed down due to low temperature. So, the development of rancidity due to oxidation is retarded.

4. Rancidity can be retarded by storing food in air-tight containers:

When food is stored in air-tight containers, then there is little exposure to oxygen of air. Due to reduced exposure to oxygen, the oxidation of fats and oils present in food is slowed down and hence the development of rancidity is retarded.

5. Rancidity can be retarded by storing foods away from light:

In the absence of light, the oxidation of fats and oils present in food is slowed down and hence the development of rancidity is retarded.

Example: Can rancidity retarded by storing foods away from light?

Solution: In the absence of light, the oxidation of fats and oils present in food is slowed down and hence the development of rancidity is retarded.

Example: A silver article generally turns black when kept in the open for few days. The article when rubbed with toothpaste it regain its shining.

(a) Why do silver article turn black

(b) Name the black substance formed with its molecular formula

(c) Why it again starts shining by rubbing with toothpaste?

Solution : (a) Silver reacts with air, mixture, acids, gases, etc. present in air and form undesirable compounds on their surfaces and thus it tarnishes and gets black coating

(b) Silver sulphide (Ag_2S)

(c) Toothpaste contains traces of hydrogen peroxide which reacts with Ag_2S formed on the surface and restore whiteness.

Example: Give an example where corrosion is an advantage rather than a disadvantage.

Solution: Due to corrosion, aluminium objects lose their lustre and become dull after some time. This is due to the reason that when aluminium is exposed to moist air, its surface is covered with a thin impervious layer of aluminium oxide (Al_2O_3). This layer does not allow moist air to come in contact with the metal and hence protects the metal underneath from further damage.

CONCEPT OF EQUIVALENT AND NORMALITY

(a) Equivalent mass:

Number of parts by mass of an element which reacts or displaces from a compound 1.008 parts by mass of hydrogen, 8 parts by mass of oxygen and 35.5 parts by mass of chlorine, is known as the equivalent weight of that element.

e.g. $2Mg + O_2 \longrightarrow 2MgO$

48g 32g

12g 8g

$\because$ 32 g of O_2 reacts with 48 g of Mg

$\therefore$ 8 g of $O_2 = \frac{48 \times 8}{32} = 12$ g

$\therefore$ Equivalent weight of Mg = 12

Similarly, $Zn + H_2SO_4 \longrightarrow ZnSO_4 + H_2$

65.5 g 32.75

$\therefore$ Equivalent weight of Zn = 32.75 g

$Al + \frac{3}{2} Cl_2 \longrightarrow AlCl_3$

27 g $\frac{3}{2} \times 71$ g

$\because$ 111.5 g chlorine reacts with 27 g of Al.

$\therefore$ 35.5 chlorine reacts with $\frac{27 \times 35.5}{111.5} = 9.0$ g of Al

$\therefore$ Equivalent weight of aluminium = $\frac{27}{3} = 9.0$

As we can see from the above examples that equivalent weight is the ratio of atomic weight and a factor (say n-factor or valency factor) which is in above three cases is their respective valencies.

Equivalent weight (E) = $\frac{\text{Atomic weight or Molecular weight}}{\text{Valency factor (v.f.)}} = \frac{\text{Mol. wt.}}{\text{n - factor}} = \frac{M}{x}$

Number of Equivalents = $\frac{\text{mass of species}}{\text{eq. wt. of that species}}$

For a solution, Number of equivalents = N_1V_1, where N is the normality and V is the volume in litres

- Equivalent mass is a pure number which, when expressed in gram, is called gram equivalent mass.
- The equivalent mass of substance may have different values under different conditions.
- There in no hard and fast rule that equivalent weight will be always less than the molecular mass.

Valency factor calculation:

(i) For Elements:

Valency factor = valency of the element.

(ii) For Acids:

Valency factor = number of replaceable H^+ ions per acid molecule

Example: What is the relation between the molecular weight & equivalent wt of the H_2SO_4 for given reaction:

$NaOH + H_2SO_4 \longrightarrow NaHSO_4 + H_2O$

Base Acid

Solution: Valency factor of acid = 1

Here, one of molecule of H_2SO_4 replaced one OH^- from NaOH. Therefore, valency factor for H_2SO_4 is one

$\therefore$ Eq. wt. of $H_2SO_4 = \frac{\text{Mol.wt}}{1}$

Example: What is the relation between the molecular weight & equivalent wt of the following:

NaOH, KOH

Solution: v.f. $\rightarrow$ 1 1

Eq. wt. $\rightarrow$ $\frac{M}{1}$ $\frac{M}{1}$

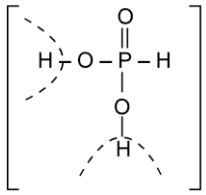
- Bases may be defined as the substances in which OH group is/are directly attached with group I elements (Li, Na, K, Rb, Cs), group II elements (Be, Mg, Ca, Ba) or group III elements (Al, Ga, In, Tl), transition metals, non-metallic cations like PH_4^+, NH_4^+ etc.

(iv) For Bases in Acid - base reactions:

In case of acid base reaction, the valence factor is the actual number of H^+ or OH^- replaced in the reaction. The acid or base may contain more number of replaceable H^+ or OH^- than actually replaced in reaction.

- v. f. for base is the number of H^+ ion from the acid replaced by each molecule of the base.

Example: What is the relation between the molecular weight & equivalent wt of the following:

HCl, H_2SO_4 H_3PO_4 H_3PO_3 

{see there are only two replaceable H^+ions}

Solution: Valency factor $\rightarrow$ 1 2 3 2

(assume 100% dissociation)

Eq. wt. (E) $\rightarrow$ $\frac{M}{1}$ $\frac{M}{2}$ $\frac{M}{3}$ $\frac{M}{2}$

- Replaceable hydrogen atoms are those hydrogen atoms which are attached with the atoms of group VI and group VII i.e. O, S, Se, Te, & F, Cl, Br, I.

(iii) For Bases:

Valency factor = number of replaceable OH^- ions per base molecule.

Example: Find out the valency factor of acid & base in the given reaction:

$2NaOH + H_2SO_4 \longrightarrow Na_2SO_4 + 2H_2O$

Base Acid

Solution: Valency factor of base = 1

Here, two molecule of NaOH replaced $2H^+$ ion from the H_2SO_4. Therefore, each molecule of NaOH replaced only one H^+ ion of acid, so v.f. = 1.

- v. f. for acid is the number of OH^- replaced from the base by each molecule of acid.

(v) Salts:

(a) In non-reacting condition

- Valency factor = Total number of positive charge or negative charge present in the compound.

Example: Calculate the valancy factor of following: Na_2CO_3, $Fe_2(SO_4)_3$, $FeSO_4.7H_2O$

Solution:	Na_2CO_3,	$Fe_2(SO_4)_3$	$FeSO_4.7H_2O$
V.f.	2	$2\times3 = 6$	2
Eq.wt.	$\frac{M}{2}$	$\frac{M}{6}$	$\frac{M}{2}$

Mind it: In case of hydrated salt, positive/negative charge of water molecule is not counted.

(b) In reaction condition

Example : Find out the valency factor for Na_2CO_3 in the given reaction:

$Na_2CO_3 + HCl \longrightarrow NaHCO_3 + NaCl$

Base Acid

Solution : It is an acid base reaction, therefore valency factor for Na_2CO_3 is one while in non-reacting condition, it will be two.

(vi) Equivalent weight of oxidising / reducing agents in a redox reaction

In case of redox change,

v.f. = Total change in oxidation number per molecule.

$$\text{Equivalent weight of oxidising / reducing agent} = \frac{\text{Molecular Mass}}{\text{Change in oxidation Number}}$$

Example: What is the relation between the molecular weight & equivalent wt of the $KMnO_4$ for given reaction:

$KMnO_4 + H_2O_2 \longrightarrow Mn^{2+} + O_2$

Solution: Mn in $KMnO_4$ is going from +7 to +2, so change in oxidation number per molecule of $KMnO_4$ is 5. So, the valency factor of $KMnO_4$ is 5 and equivalent weight is $\frac{M}{5}$.

Normality:

Normality of a solution is defined as the number of equivalents of solute present in one litre (1000 mL) solution.

Let V mL of a solution is prepared by dissolving W g of solute of equivalent weight E in water.

- Number of equivalents of solute = $\frac{W}{E}$

VmL of solution contain $\frac{W}{E}$ equivalents of solute

$\therefore$ 1000 mL solution will contain $\frac{W \times 1000}{E \times V}$ equivalents of solute.

- **Normality (N)** = $\frac{W \times 1000}{E \times V}$
- **Normality (N) = Molarity × Valency factor**

  N × V (in mL) = M × V (in mL) × n

  or
- **milliequivalents = millimoles × n**

Relation between Molarity and Normality

S = Molarity × molecular mass of solute

and S = Normality × equivalent mass of solute.

So we can write

Molarity × Molecular mass of solute = Normality × Equivalent mass of solute.

$$\text{Normality} = \frac{\text{Molarity} \times \text{Molecular mass of solute}}{\text{Equivalent mass of solute}}$$

$$= \frac{\text{Molarity} \times \text{Molecular mass of solute}}{\left(\text{Molecular mass of solute} / \text{Valency factor}\right)}$$

Normality = Molarity × Valency factor

$N = M \times n, \quad N \geq M$

Example: Calculate the normality of a solution containing 15.8 g of $KMnO_4$ in 50 mL acidic solution.

Solution: **Normality (N)** = $\frac{W \times 1000}{E \times V}$

Here W = 15.8 g, V = 50 mL

$E = \frac{\text{molar mass of } KMnO_4}{\text{Valency factor}} = 158/5 = 31.6$ So, normality = 10 N

Example: Calculate the normality of a solution containing 50 mL of 5 M solution of $K_2Cr_2O_7$ in acidic medium.

Solution: Normality (N) = Molarity × valency factor = 5 × 6 = 30 N

(b) Law of equivalence

The law states that one equivalent of an element combine with one equivalent of the other. In a chemical reaction, equivalents and milli equivalents of reactants react in equal amount to give same number of equivalents or milli equivalents of products separately.

Accordingly

(i) $a_A + b_B \rightarrow m_M + n_N$

meq of A = meq of B = meq of M = m.eq. of N

(ii) In a compound M_xN_y

meq of M_xN_y = meq of M = meq of N

Example: What mass of MnO_2 is reduced by 35 mL of 0.16 N oxalic acid in acid solution?

$MnO_2 + H^+ + H_2C_2O_4 \longrightarrow Mn^{2+} + CO_2 + H_2O$

(A) 8.7 g (B) 0.25 g (C) 0.84g (D) 43.5 g

Solution: (B) equivalents of oxalic acid = equivalents of MnO_2

= 35 mL of 0.16 N

wt. of MnO_2 = ENV/1000 = 45 × 0.16 × 35/1000 = 0.252g

Example: Find the number of moles of $KMnO_4$ needed to oxidise one mole Cu_2S in acidic medium.

The reaction is $KMnO_4 + Cu_2S \longrightarrow Mn^{2+} + Cu^{2+} + SO_2$

Solution: From law of equivalence,

equivalents of Cu_2S = equivalents of $KMnO_4$

moles of Cu_2S × v.f. = moles of $kMnO_4$ × v.f.

1 × 8 = moles of $KMnO_4$ × 5 $\Rightarrow$ **moles of $KMnO_4$ = 8/5**

($\therefore$ v.f. of Cu_2S = 2 (2 – 1) + 1 (4 – (–2)) = 8 and v.f. of $KMnO_4$ = 1 (7 –2) = 5)

Example: The number of moles of oxalate ions oxidized by one mole of MnO_4^- ion in acidic medium are:

(A) $\frac{5}{2}$ (B) $\frac{2}{5}$ (C) $\frac{3}{5}$ (D) $\frac{5}{3}$

Solution: Equivalents of $C_2O_4^{2-}$ = equivalents of MnO_4^-

x(mole) × 2 = 1 × 5

($\therefore$ v.f. of $C_2O_4^{2-}$ = 2 (4 – 3) = 2

and v.f. of MnO_4^- = 1 (7 – 2) = 5).

x = $\frac{5}{2}$ mole of $C_2O_4^{2-}$ ions.

Example: How many milliliters of 0.02 M $KMnO_4$ solution would be required to exactly titrate 25 mL of 0.2 M $Fe(NO_3)_2$ solution in acidic medium?

Solution: Method -1: Mole concept method

Starting with 25 mL of 0.2 M Fe^{2+}, we can write:

Millimoles of Fe^{2+} = 25 × 0.2 .....(1)

and in volume V (in milliliters) of the $KMnO_4$,

Millimoles of MnO_4^- = V (0.02) .....(2)

The balanced reaction is:

$MnO_4^- + 5Fe^{2+} + 8H^+ \longrightarrow Mn^{2+} + 5Fe^{3+} + 4H_2O$

This requires that at the equivalent point,

$$\frac{\text{m.moles of } MnO_4^-}{1} = \frac{\text{m.moles of } Fe^{2+}}{5}$$

$\therefore \quad \frac{V(0.02)}{1} = \frac{(25)(0.2)}{5}$ (from (1) & (2)

$\therefore$ V = 50 mL.

Method -2: Equivalent Method:

At the equivalence point,

milliequivalents of MnO_4^- = milliequivalents of Fe^{2+}

$M_1 \times Vf_1 \times V_1 = M_2 \times Vf_2 \times V_2$

$0.02 \times 5 \times V_1 = 0.2 \times 1 \times 25$

($\because$ $MnO_4^- \rightarrow Mn^{2+}$; v.f. = 5, $Fe^{2+} \rightarrow Fe^{3+}$; v.f. = 1)

$\therefore$ V_1 = 50 mL.

TITRATION

Titration is a procedure for determining the concentration of a solution by allowing a carefully measured volume to react with a standard solution of another substance, whose concentration is known.

Standard solution - It is a solution whose concentration is known and is taken in burette. It is also called **Titrant.**

There are two type of titrants:

- **Primary titrants/standard -** These reagents can be accurately weighed and their solutions are not to be standardised before use.
  Ex: Oxalic acid, $K_2Cr_2O_7$, $AgNO_3$, $CuSO_4$, ferrous ammonium sulphate, hypo etc.
- **Secondary titrants/standard:** These reagents cannot be accurately weighed and their solutions are to be standardised before use.
  Ex: NaOH, KOH, HCl, H_2SO_4, I_2, $KMnO_4$ etc.

Titrate: Solution consisting of substance to be estimated, generally taken in a beaker .

Equivalence point: It is the point when number of equivalents of titrant added becomes equal to number of equivalents of titrate.

At equivalence point:

$n_1V_1M_1 = n_2V_2M_2$

Indicator: An auxiliary substance added for physical detection of completion of titration at equivalence point. It generally show colour change on completion of titration.

Type of Titrations:

- **Acid-base titrations (to be studied in Ionic equilibrium)**

- **Redox Titrations**

Some Common Redox Titrations

Table of Redox Titrations: (Excluding Iodometric / Iodimetric titrations)

	Estimation of	By titrating with	Reactions	Relation*between OA and RA
1.	Fe^{2+}	MnO_4^-	$Fe^{2+} \longrightarrow Fe^{3+} + e^-$ $MnO_4^- + 8H^+ + 5e^- \longrightarrow Mn^{2+} + 4H_2O$	$5Fe^{2+} \equiv MnO_4^-$ Eq. wt. of Fe^{2+} = M/1
2.	Fe^{2+}	$Cr_2O_7^{2-}$	$Fe^{2+} \longrightarrow Fe^{3+} + e^-$ $Cr_2O_7^{2-} + 14H^+ + 6e^- \longrightarrow 2Cr^{3+} + 7H_2O$	$5Fe^{2+} \equiv Cr_2O_7^{2}$ Eq.wt. of $Cr_2O_7^{2-}$ = M/6
3.	$C_2O_4^{2-}$	MnO_4^-	$C_2O_4^{2-} \longrightarrow 2CO_2 + 2e^-$ $MnO_4^- + 8H^+ + 5e^- \longrightarrow Mn^{2+} + 4H_2O$	$5C_2O_4^{2-} \equiv 2MnO_4^-$ Eq. wt. of $C_2O_4^{2-}$ = M/2
4.	H_2O_2	MnO_4^-	$H_2O_2 \longrightarrow 2H^+ + O_2 + 2e^-$ $MnO^- + 8H^+ + 5e^- \longrightarrow Mn^{2+} + 4H_2O$	$5H_2O_2 \equiv 2MnO_4^-$ Eq.wt. of H_2O_2 = M/2
5.	As_2O_3	MnO_4^-	$As_2O_3 + 5H_2O \longrightarrow 2AsO_4^{3-} + 10H^+ + 4e^-$ $MnO_4^- + 8H^+ + 5e^- \longrightarrow Mn^{2+} + 4H_2O$	Eq. wt. of As_2O_3 = M/4
6.	AsO_3^{3-}	BrO_3^-	$AsO_3^{3-} + H_2O \longrightarrow AsO_4^{3-} + 2H^+ + 2e^-$ $BrO_3^- + 6H^+ + 6e^- \longrightarrow Br^- + 3H_2O$	Eq. wt. of AsO_3^{3-} = M/2 Eq. wt. of BrO_3^- = M/6

Permanganate Titrations:

- $KMnO_4$ is generally used as oxidising agent in acidic medium, generally provided by dilute H_2SO_4.
- $KMnO_4$ works as self indicator persistent pink color is indication of end point.
- Mainly used for estimation of Fe^{2+}, oxalic acid, oxalates, H_2O_2 etc.

Example: Write the balanced reaction of titration of $KMnO_4$ vs ferrous ammonium sulphate in presence of H_2SO_4.

Solution: **Reaction:** $2KMnO_4 + 10[FeSO_4(NH_4)2SO_4.\ 6H_2O] + 8H_2SO_4 \longrightarrow$

$5Fe_2(SO_4)_3 + 10(NH_4)_2SO_4 + K_2SO_4 + 2MnSO_4 + 68H_2O$

Redox Changes: $Fe^{2+} \longrightarrow Fe^{3+} + e^- \left(E_{FeSO_4} = \frac{M}{1}\right)$

$Mn^{7+} + 5e^- \longrightarrow Mn^{2+} \left(E_{KMnO_4} = \frac{M}{5}\right)$

Indicator: $KMnO_4$ acts as self indicator

Example: Write the balanced reaction of titration of $KMnO_4$ vs oxalic acid in presence of H_2SO_4.

Solution: Reaction: $2KMnO_4 + 3H_2SO_4 + 5H_2C_2O_4 \longrightarrow K_2SO_4 + 2MnSO_4 + 8H_2O + 10CO_2$

Redox Changes: $C_2^{3+} \longrightarrow 2C^{4+} + 2e^- \left(E_{H_2C_2O_4} = \frac{M}{2}\right)$

$5e + Mn^{7+} \longrightarrow Mn^{2+} \left(E_{KMnO_4} = \frac{M}{5}\right)$

Indicator: $KMnO_4$ acts as self indicator.

Volume Strength of H_2O_2 Solution

When a solution of H_2O_2 is labeled as 'x volume', it means that 1 volume (1 ml or 1 litre) of H_2O_2 solution would liberate x volumes (1 ml or 1 litre) of O_2 at STP on complete decomposition.

$H_2O_2 \xrightarrow{\Delta} H_2O + ½O_2$ …….. (i)

If a H_2O_2 solution (acting as reducing agent) has normality N and its is to be reacted with $KMnO_4$ solution (acting as oxidizing agent). Our task is to determine its volume strength. We can say that there are N equivalents of H_2O_2 present in 1 litre of this H_2O_2 solution.

1 ml of H_2O_2 of this solution would contain $\frac{N}{1000}$ equivalents.

$H_2O_2 + KMnO_4 \xrightarrow{H^+} O_2 + Mn^{+2}$ …….. (ii)

Moles of H_2O_2 in 1 ml of this solution = $\frac{1}{2}\times\frac{N}{1000}$ [from equation (ii)]

When these many moles of H_2O_2 in 1 ml of solution are allowed to decompose according to the reaction, $H_2O_2 \xrightarrow{\Delta} H_2O + ½O_2$, the volume of O_2 released (in ml) by them at STP will give the volume strength of H_2O_2 solution.

Moles of O_2 given by 1 ml of this solution = $\frac{1}{2}\times\frac{1}{2}\times\frac{N}{1000}$ [from equation (i)]

Volume of O_2 at STP given by 1ml of this solution = $\frac{1}{2}\times\frac{1}{2}\times\frac{N}{1000} \times 22400 = 5.6 \times N$

Volume strength of H_2O_2 = 5.6 × Normality of H_2O_2

Volume strength of H_2O_2 = 11.2 × Molarity of H_2O_2

Parts per million (ppm)

When the solute is present in very less amount, then this concentration term is used. It is defined as the number of parts of the solute present in every 1 million parts of the solution. ppm can both be in terms of mass or in terms of moles. If nothing has been specified, we take ppm to be in terms of mass. Hence, a 100 ppm solution means that 100 g of solute is present in every 1000000 g of solution.

$ppm_A = \frac{\text{mass of A}}{\text{Total mass}}\times 10^6 = \text{mass fraction} \times 10^6$

Measurement of Hardness:

Hardness of water is a characteristics property by which water 'prevent lathering of soap'.

Causes: The hardness of water is due to presence of certain salts (mainly bicarbonates, sulphates and chlorides) of Ca, Mg and few other heavy metal salts dissolved in water.

A sample of hard water when treated with soap (sodium or potassium salt of higher fatty acid like oleic, palmitic or stearic) do not produce lather with soap, but on the other hand forms a white scum or precipitate.

The precipitate is formed due to insoluble soaps of Ca and Mg.

Reactions:-

$2C_{17}H_{35}COONa + CaCl_2 \rightarrow Ca(C_{17}H_{35}COO)_2\downarrow + NaCl$

Sodium Stearate (Sodium soap) hardness Calcium Stearate (Insoluble)

$2\ C_{17}H_{35}COONa + MgSO_4 \rightarrow (C_{17}H_{35}COO)_2Mg\downarrow + Na_2SO_4$

Magnesium Stearate (Insoluble)

Thus the water which does not produce lather with soap solution readily is called **HARD WATER** and water which lathers easily on shaking with soap solution is called **SOFT WATER.**

Types of Hardness of Water:

Hardness of water is of two types:

(a) Temporary hardness.

(b) Permanent hardness.

(a) Temporary Hardness:

Temporary hardness is due to the presence of bicarbonates of calcium and magnesium.

(i) It can be removed by boiling. On boiling the carbon dioxide is expelled out of water and precipitates of calcium carbonate and magnesium carbonate are deposited at the bottom.

$Ca(HCO_3)_2 \xrightarrow{\text{Boiling}} CaCO_3 + H_2O + CO_2$

$Mg(HCO_3)_2 \xrightarrow{\text{Boiling}} MgCO_3 + H_2O + CO_2$

Boiling is an expensive method of removing hardness of water on a large scale.

(ii) Temporary hardness can also be removed by adding lime or calcium hydroxide (Clark's process) to water. Lime absorbs carbon dioxide and precipitates of calcium carbonate are formed which settle at the bottom and separated by filtration.

$Ca(HCO_3)_2 + Ca(OH)_2 — 2CaCO_3 + 2H_2O.$

(b) Permanent Hardness:

Permanent hardness is due to the presence of chlorides and sulphates of calcium and magnesium. It is not removed by boiling. However it can be removed by following methods:

(i) When sodium carbonate or soda ash is added to water, it removes both temporary and permanent hardness of water. The sulphate of calcium and magnesium is converted into sodium sulphate.

$CaSO_4 + Na_2CO_3 \longrightarrow Na_2SO_4 + CaCO_3$

Total Hardness = Temporary Hardness + Permanent Hardness.

Measurement of Hardness:

The concentrations of hardness as well as non-hardness constituting ions are usually expressed in terms of equivalent amount of $CaCO_3$.

The choice of $CaCO_3$ in particular is due to its molecular weight 100 (equivalent weight = 50) and it is the most insoluble salt that can be precipitated in water treatment.

Hardness of the hardness causing salt in terms of $CaCO_3$.

The equivalents of $CaCO_3$

$$= \frac{\text{Mass of hardness causing substance x chemical equivalent of } CaCO_3 \times \text{mol. wt of } CaCO_3}{\text{Chemical equivalent or molecular Wt of hardness causing substance}}$$

$$= \frac{\text{Mass of hardness causing substance} \times 50 \times 100}{\text{Chemical equivalent or molecular Wt of hardness causing substance}}$$

Hardness is measured in terms of ppm (parts per million) of $CaCO_3$ or equivalent to it.

$$\text{Hardness in ppm} = \frac{\text{mass of } CaCO_3}{\text{Total mass of solution}} \times 10^6$$

Units of hardness:

1. Parts Per Million (ppm): is the parts of $CaCO_3$ equivalent hardness per 10^6 parts of water i.e., 1ppm= 1 part of $CaCO_3$ equivalent hardness in 10^6 parts of water.
2. Milligrams Per Litre (mg/L): number of milligrams of $CaCO_3$ equivalent hardness present per liter of water.

   1mg/L = 1mg of $CaCO_3$ equivalent hardness of 1L of water = 1kg = 1000g = 10^6mg.

∴ 1mg/L = 1mg of $CaCO_3$ eq per 10^6 mg of water = 1ppm.

Example: How many grams of $MgCO_3$ dissolved per litre gives 84 ppm hardness?

(A) 70.56 mg/L (B) 48.23 mg/L (C) 81.49mg/L (D) 66.12 mg/L

Solution: (A)

Example: 0.00012% $MgSO_4$ and 0.000111% $CaCl_2$ is present in water. What is the measured hardness of water and millimoles of washing soda required to purify water 1000 L water?

Solution: Basis of calculation = 100 g hard water

$MgSO_4 = 0.00012g = \frac{0.00012}{120}mole$

$CaCl_2 = 0.000111g = \frac{0.000111}{111}mole$

$\therefore$ equivalent moles of $CaCO_3 = \left(\frac{0.00012}{120}+\frac{0.000111}{111}\right)mole$

$\therefore$ mass of $CaCO_3 = \left(\frac{0.00012}{120}+\frac{0.000111}{111}\right)\times 100 = 2 \times 10^{-4}$ g

Hardness (in terms of ppm of $CaCO_3$) = $\frac{2\times10^{-4}}{100}\times10^{6} = 2$ ppm

$CaCl_2 + Na_2CO_3 \longrightarrow CaCO_3 + 2NaCl$

$MgSO_4 + Na_2CO_3 \longrightarrow MgCO_3 + Na_2SO_4$

$\therefore$ Required Na_2CO_3 for 100g of water = $\left(\frac{0.00012}{120}+\frac{0.000111}{111}\right)mole = 2 \times 10^{-6}$ mole

$\therefore$ Required Na_2CO_3 for 1000 litre water = $\frac{2\times10^{-6}}{100}\times10^{6} = \frac{2}{100}mole\ (\because$ d = 1g/mL)

$= \frac{20}{1000}mole = 20$ millimole

EXERCISE-1

General Introduction of Chemistry & Characteristics of Chemical Reactions

1. Which of the following statements is/are true?

(A) The total mass of the system remains same in a chemical change

(B) A chemical change is permanent and irreversible

(C) A physical change is temporary and reversible

(D) All of these

2. Which of the following statements is/are correct?

(A) A chemical equation tells us about the substances involved in a reaction.

(B) A chemical equation informs us about the symbols and formulae of the substances involved in a reaction.

(C) A chemical equation tells us about the atoms or molecules of the reactants and products involved in a reaction.

(D) All are correct

3. Match the column-I with column-II and select the correct option from the codes given below:

Column-I		Column-II	
a.	Expansion of metals on heating.	i.	Neither physical nor chemical change
b.	A stone kept in the sunlight	ii.	Chemical change
c.	Burning of a candle	iii.	Combination of physical and chemical changes
d.	Curdling of milk	iv.	Physical change

(A) a-iv, b-i, c-iii, d-ii

(B) a-iv, b-iii, c-ii, d-i

(C) a-i, b-ii, c-iii, d-iv

(D) a-i, b-iv, c-ii, d-iii

4. Select the correct statement among the following.

(A) Crystallization is a chemical change

(B) Digestion of food is a chemical change

(C) Burning of paper is a temporary change

(D) A reversible change is always chemical

5. Read the given statements and select the correct option.

Statement-1: Breaking of a bone china plate is a physical change

Statement-2: When a bone China plate breaks, the pieces can be joined to get back the original plate.

(A) Both statements 1 and 2 are true and statement-2 is the correct explanation statement-1.

(B) Both statements 1 and 2 are true but statement-2 is not the correct explanation statement-1.

(C) Statement-1 is true and statement-2 is false.

(D) Both Statement-1 is true and statement-2 is false.

6. The reaction between aqueous solutions of sodium chloride and silver nitrate is –

(A) displacement reaction

(B) synthesis reaction

(C) double displacement reaction

(D) analysis reaction

7. $CH_4 + Cl_2 \longrightarrow CH_3Cl + HCl$ reaction is an example of -

(A) Synthetic (B) Analytical

(C) Displacement (D) Neutralisation

8. A Brown and bright element "x" when heated in presence of air turns into black substance "y". If Hydrogen gas is passed over this heating material again "x" is obtained. "x" and "y" are–

(A) Cu & CuO (B) S & SO_2

(C) C & CO_2 (D) Na & NaH

9. Which of the following is endothermic reaction?

(A) $C(s) + O_2(g) \longrightarrow CO_2(g)$

(B) $N_2(g) + O_2(g) \longrightarrow 2NO(g)$

(C) $2H_2(g) + O_2(g) \longrightarrow 2H_2O(l)$

(D) $2CH_3OH(l) + 3O_2(g) \longrightarrow 2CO_2(g) + 4H_2O(l)$

10. When a burning splinter is brought near the gas jar containing hydrogen gas a popping sound is observed. It is _____ reaction.

(A) exothermic

(B) endothermic

(C) exothermic and endothermic

(D) none of these

Chemical Equations & Balancing

1. In the balanced equation -

$aFe_2O_3 + bH_2 \longrightarrow cFe + dH_2O$

The values of a, b, c, d are respectively -

(A) 1, 1, 2, 3 (B) 1, 1, 1, 1

(C) 1, 3, 2, 3 (D) 1, 2, 2, 3

2. Which of the following reactions is not balanced?

(A) $2NaHCO_3 \longrightarrow Na_2CO_3 + H_2O + CO_2$

(B) $2C_4H_{10} + 12O_2 \longrightarrow 8CO_2 + 10H_2O$

(C) $2Al + 6H_2O \longrightarrow 2Al(OH)_3 + 3H_2$

(D) $4NH_3 + 5O_2 \longrightarrow 4NO + 6H_2O$

3. The equation - $Cu + xHNO_3 \rightarrow Cu(NO_3)_2 + yNO_2 + 2H_2O$

The values of x and y are -

(A) 3 and 5 (B) 8 and 6

(C) 4 and 2 (D) 7 and 1

4. In the reaction $FeSO_4 + x \rightarrow Na_2SO_4 + Fe(OH)_2$, x is –

(A) Na_2SO_4 (B) H_2SO_4

(C) NaOH (D) none of these

5. In the reaction $xPb(NO_3)_2 \rightarrow yPbO + zNO_2 + O_2$ x, y and z are -

(A) 1,2,2 (B) 2,2,4

(C) 1,2,4 (D) 4,2,2

6. In balancing the reaction.

$xH_2S + 2NaNO_3 + 2HCl \rightarrow y\,S + zNO + kNaCl + 4H_2O$

one would get x, y, z and k, respectively, as

(A) 3, 3, 2 and 2 (B) 2, 2, 3 and 3

(C) 3, 3, 4 and 4 (D) 4, 4, 3 and 3

Redox Reactions & Oxidation Number

1. In the reaction $Mg + Cl_2 \rightarrow MgCl_2$ Magnesium may be regarded as -

(A) an oxidizing agent

(B) a reducing agent

(C) a catalyst

(D) providing an inert medium

2. When the gases sulphur dioxide and hydrogen sulphide react, the reaction is

$SO_2 + 2H_2S \rightarrow 2H_2O + 3S$

Here hydrogen sulphide is acting as -

(A) an oxidizing agent (B) a reducing agent

(C) a dehydrating agent (D) a catalyst

3. Which of the following statements is correct for oxidation reaction?

(A) Gain or addition of electronegative radical or element

(B) Removal of hydrogen atom

(C) Removal or loss of electropositive radical or element

(D) All the above statements are correct.

4. $CuO + H_2 \longrightarrow H_2O + Cu$, reaction is an example of -

(A) redox reaction

(B) synthesis reaction

(C) neutralisation

(D) analysis reaction

5. Which of the following is an example of oxidation reaction?

(A) $Sn^{+2} - 2e^- \rightarrow Sn^{+4}$ (B) $Fe^{+3} + e^- \rightarrow Fe^{+2}$

(C) $Cl_2 + 2e^- \rightarrow 2Cl^-$ (D) None of these

6. In the process of burning of magnesium in air, magnesium undergoes -

(A) reduction (B) sublimation

(C) oxidation (D) all of these

7. A substance which oxidizes itself and reduces other is known as -
(A) an oxidizing agent (B) a reducing agent
(C) Both of these (D) None of these

8. Oxidation is a process which involves -
(A) addition of oxygen
(B) removal of hydrogen
(C) loss of electrons
(D) All are correct

9. In the reaction $PbO + C \rightarrow Pb + CO$.
(A) PbO is oxidized
(B) C acts as oxidizing agent
(C) C acts as a reducing agent
(D) This reaction does not represent a redox reaction

10. A redox reaction is one in which -
(A) both the substances are reduced
(B) both the substances are oxidized
(C) an acid is neutralised by the base
(D) one substance is oxidized, while the other is reduced

11. Which of the following is not an example of redox reaction?
(A) $CuO + H_2 \longrightarrow Cu + H_2O$
(B) $Fe_2O_3 + 3CO \longrightarrow 2Fe + 3CO_2$
(C) $2K + F_2 \longrightarrow 2KF$
(D) $BaCl_2 + H_2SO_4 \longrightarrow BaSO_4 + 2HCl$

12. When $KMnO_4$ acts as an oxidizing agent and ultimately forms MnO_4^{2-}, MnO_2, Mn_2O_3 and Mn^{2+}, then the number of electrons transferred in each case is:
(A) 4, 3, 1, 5 (B) 1, 5, 3, 7
(C) 1, 3, 4, 5 (D) 3, 5, 7, 1

Types of Redox Reactions

1. Neutralization reaction is an example of -
(A) exothermic reaction
(B) endothermic reaction
(C) oxidation reaction
(D) none of these

2. $Zn(s) + H_2SO_4(aq) \longrightarrow ZnSO_4(aq) + H_2(g)$ is an example of -
(A) precipitation reaction
(B) endothermic reaction
(C) evolution of gas
(D) change in colour

3. When dilute hydrochloric acid is added to iron fillings -
(A) hydrogen gas and ferric chloride are produced
(B) chlorine gas and ferric hydroxide are produced
(C) no reaction takes place
(D) iron salt and water are produced

4. Chemical reaction $2Na + Cl_2 \longrightarrow 2NaCl$ is an example of -
(A) combination reaction
(B) decomposition reaction
(C) displacement reaction
(D) double displacement reaction

5. Which of the following equations is representing combination of two elements?
(A) $CaO + CO_2 \longrightarrow CaCO_3$
(B) $4Na + O_2 \longrightarrow 2Na_2O$
(C) $SO_2 + 1/2\ O_2 \longrightarrow SO_3$
(D) $2Na + 2H_2O \longrightarrow 2NaOH + H_2$

6. Which of the following equations is not an example of single displacement reaction?
(A) $2Al + Fe_2O_3 \longrightarrow Al_2O_3 + 2Fe$
(B) $Ca + Cl_2 \longrightarrow CaCl_2$
(C) $2KI + Cl_2 \longrightarrow 2KCl + I_2$
(D) $2Na + 2H_2O \longrightarrow 2NaOH + H_2$

7. Which of the following is/are a decomposition reaction(s)?
(A) $2HgO \longrightarrow 2Hg + O_2$
(B) $CaCO_3 \longrightarrow CaO + CO_2$
(C) $2H_2O \longrightarrow 2H_2 + O_2$
(D) All of these

8. Match the following -

Column-A Types of chemical reaction		Column-B Chemical	
(a)	Combination reaction	(i)	$CaCO_3 \longrightarrow CaO + CO_2$
(b)	Decomposition reaction	(ii)	$2H_2O \longrightarrow 2H_2 + O_2$
(c)	Displacement reaction	(iii)	$CaO + CO_2 \longrightarrow CaCO_3$
(d)	Analysis reaction	(iv)	$Fe(s) + CuSO_4$ (aq.) $\longrightarrow FeSO_4$ (aq) + Cu(s)

(A) a (ii), b (i), c (iv), d (iii)
(B) a (i), b (ii), c (iii), d(iv)
(C) a (iii), b (i), c (iv), d (ii)
(D) a (iii), b (i), c (iii), d (iv)

9. Which of the following reactions is/are a double displacement reaction(s)?
(i) $AgNO_3 + NaBr \longrightarrow NaNO_3 + AgBr$
(ii) $BaCl_2 + H_2SO_4 \longrightarrow BaSO_4 + 2HCl$
(iii) $As_2O_3 + 3H_2S \longrightarrow As_2S_3 + 3H_2O$
(iv) $NaOH + HCl \longrightarrow NaCl + H_2O$
(A) (i) & (ii) (B) only (iii)
(C) only (iv) (D) (i) to (iv) all

10. $AgNO_3$ (aq) + NaCl (aq) $\longrightarrow$ AgCl(s) + $NaNO_3$ (aq)
Above reaction is a -
(A) precipitation reaction
(B) double displacement reaction
(C) combination reaction
(D) (A) and (B) both

11. $H_2SO_4 + 2NaOH \longrightarrow Na_2SO_4 + 2H_2O$
Above equation is a -
(i) neutralization reaction
(ii) double displacement reaction
(iii) decomposition reaction
(iv) addition reaction
(A) (i) to (iv) all (B) (i) and (ii)
(C) (i) and (iii) (D) (ii) and (iv)

12. $Zn + H_2SO_4$ (dil) $\longrightarrow ZnSO_4 + H_2 \uparrow$
Above equation is a -
(A) decomposition reaction
(B) single displacement reaction
(C) combination reaction
(D) synthesis reaction

13. The reaction in which two compounds exchange their ions to form two new compounds is -
(A) a displacement reaction
(B) a decomposition reaction
(C) an addition reaction
(D) a double displacement reaction

14. Stainless steel form by mixing iron, chromium and nickel is a -
(A) a displacement reaction
(B) a decomposition reaction
(C) an oxidation reaction
(D) None of these

15. Burning of fuel is an example of -
(A) Redox reaction
(B) displacement reaction
(C) combustion reaction
(D) (A) & (C)

16. $H_2S(g) + Cl_2$ (g) $\longrightarrow$ 2 HCl(g) + S(s), The reaction is interpreted as:
(A) H_2S is getting oxidized and Cl_2 is getting reduced
(B) H_2S is getting reduced and Cl_2 is getting oxidized
(C) Only H_2S is oxidized
(D) Both H_2S and Cl_2 are reduced

17. Oxidation is defined as:
(A) loss of electron (B) gain of electron
(C) loss of proton (D) gain of proton

18. The chemical reaction $HNO_3 + KOH \longrightarrow KNO_3 + H_2O$ is an example of
(A) neutralization
(B) double displacement
(C) neutralization and double displacement
(D) combination

Application of Redox Reaction

1. The term 'rancidity' represents -
(A) acid rain
(B) oxidation of fatty food
(C) rotting of fruit
(D) fading of coloured clothes in the sun.

2. Combination of phosphorus and oxygen is an example of -
(A) oxidation (B) reduction
(C) rancidity (D) none of these

3. Rancidity can be prevented by -
(A) adding antioxidants
(B) packaging oily food in nitrogen gas
(C) both (A) and (B)
(D) none of these

4. Which of the following is a combustion reaction?
(A) Boiling of water
(B) Melting of wax
(C) Burning of petrol
(D) None of these

5. Nitrogen gas is filled in packets of chips because:
(A) It is unreactive
(B) It is major part of atmosphere
(C) Air contains oxygen
(D) None of these

Oxidation & Reduction

1. The oxidation number of Oxygen in Na_2O_2 is:
(A) + 1 (B) + 2
(C) – 2 (D) – 1

2. One of the following has both positive and negative oxidation states
(A) F (B) Cl
(C) He (D) Na

3. The oxidation state of osmium (Os) in OsO_4 is
(A) + 7 (B) + 6
(C) + 4 (D) + 8

4. Oxidation number of nitrogen in $(NH_4)_2SO_4$ is
(A) $-\frac{1}{3}$ (B) – 1
(C) + 1 (D) – 3

5. In which of the following compounds, the oxidation number of iodine is fractional?
(A) IF_7 (B) I_3^-
(C) IF_5 (D) IF_3

6. The oxidation number of cobalt in $K_3[Co(NO_2)_6]$ is
(A) 0 (B) + 4
(C) + 3 (D) + 6

7. Phosphorus has the oxidation state of +3 in
(A) Phosphorous acid
(B) Orthophosphoric acid
(C) Hypophosphorous acid
(D) Metaphosphoric acid

8. The oxidation number of Phosphorus in $Mg_2P_2O_7$ is:
(A) + 3 (B) + 2
(C) + 5 (D) – 3

9. In which of the following compounds, nitrogen has an oxidation state of –1?
(A) N_2O (B) NO_2^-
(C) NH_2OH (D) N_2H_4

10. Arsenic estimation can be done by Bettendorff's process. The reaction is given below:
$As_4O_6 + SnCl_2 + HCl \longrightarrow As_4 + SnCl_4 + H_2O$
Find out the exact stoichiometric coefficient of the reactants respectively: (in the order as given in question)
(A) 2, 4, 6 (B) 1, 6, 12
(C) 2, 8, 20 (D) None of these

11. When white phosphorus reacts with caustic soda, the products are PH_3 and NaH_2PO_2. This reaction is an example of:
(A) Oxidation (B) Reduction
(C) Disproportionation (D) Neutralisation

12. Oxidation number of oxygen in OF_2 would be:
(A) –2 (B) +1
(C) –1 (D) +2

13. Oxidation number of Chlorine can be:
(A) From +5 to –1 (B) From +7 to –3
(C) From –7 to +1 (D) From +7 to –1

Redox Reactions

1. A reducing agent is a substance:
(A) in which an element undergoes increase in oxidation number.
(B) in which an element undergoes decrease in oxidation number.
(C) which gains electron(s)
(D) which shares electron(s)

2. Consider the following reaction:
$3Br_2 + 6CO_3^{2-} + 3H_2O \longrightarrow 5Br^- + BrO_3^- + 6HCO_3^-$
Which of the following statements is true regarding this reaction:
(A) Bromine is oxidized and the carbonate radical is reduced.
(B) Bromine is reduced and the carbonate radical is oxidized.
(C) Bromine is neither reduced nor oxidized.
(D) Bromine is both reduced and oxidized.

3. Which of the following is a redox reaction:
(A) $2\ CrO_4^{2-} + 2H^+ \rightarrow Cr_2O_7^{2-} + H_2O$
(B) $CuSO_4 + 4\ NH_3 \rightarrow [Cu\ (NH_3)_4]\ SO_4$
(C) $2Na_2\ S_2O_3 + I_2 \rightarrow Na_2S_4O_6 + 2NaI$
(D) $Cr_2O_7^{2-} + 2OH^- \rightarrow 2\ CrO_4^{2-} + H_2O$

4. Consider the reaction, $Zn + Cu^{2+} \longrightarrow Zn^{2+} + Cu$
With reference to the above, which one of the following is the correct statement?
(A) Zn is reduced to Zn^{2+}
(B) Zn is oxidised to Zn^{2+}
(C) Zn^{2+} is oxidised to Zn
(D) Cu^{2+} is oxidised to Cu

5. Which reaction does not represent auto redox or disproportionation reaction:
(A) $Cl_2 + OH^- \longrightarrow Cl^- + ClO_3^- + H_2O$
(B) $2H_2O_2 \longrightarrow H_2O + O_2$
(C) $2Cu^+ \longrightarrow Cu^{2+} + Cu$
(D) $(NH_4)_2Cr_2O_7 \longrightarrow N_2 + Cr_2O_3 + 4H_2O$

6. In the reaction $X^- + XO_3^- + H^+ \longrightarrow X_2 + H_2O$, the molar ratio in which X^- and XO_3^- react is:
(A) 1: 5 (B) 5: 1
(C) 2: 3 (D) 3: 2

7. The compound that can work both as an oxidising as well as a reducing agent is:
(A) $KMnO_4$ (B) H_2O_2
(C) $Fe_2(SO_4)_3$ (D) $K_2Cr_2O_7$

8. Which of the following behaves as both oxidising and reducing agents?
(A) H_2SO_4 (B) SO_4
(C) H_2S (D) HNO_3

9. Dichromate ion in acidic medium oxidizes stannous ion as:
$xSn^{2+} + yCr_2O_7^{2-} + zH^+ \longrightarrow aSn^{4+} + bCr^{3+} + cH_2O$
(A) the value of x: y is 1: 3
(B) the value of x + y + z is 18
(C) a: b is 3: 1
(D) the value of z – c is 9

10. The incorrect order of decreasing oxidation number of S in compounds is:
(A) $H_2S_2O_7 > Na_2S_4O_6 > Na_2S_2O_3 > S_8$
(B) $H_2SO_5 > H_2SO_3 > SCl_2 > H_2S$
(C) $SO_3 > SO_2 > H_2S > S_8$
(D) $H_2SO_4 > SO_2 > H_2S > H_2S_2O_8$

11. $K_2Cr_2O_7 + C_2O_4^{2-} + H_2SO_4 \longrightarrow K_2SO_4 + CO_2 + Cr_2(SO_4)_3 + H_2O$
In above reaction, identify the elements which do not undergo change in their oxidation state:
(A) K, C & O (B) S, H, O & Cr
(C) K, O, S & H (D) Cr & C

12. In which of the following reactions is there a change in the oxidation number of nitrogen atom:

(A) $2NO_2 \longrightarrow N_2O_4$

(B) $NH_3 + H_2O \longrightarrow NH_4^+ + OH^-$

(C) $N_2O_5 + H_2O \longrightarrow 2HNO_3$

(D) None of these

Concept of Equivalent and Titration

1. When N_2 is converted into NH_3, the equivalent weight of nitrogen will be:

(A) 1.67 (B) 2.67

(C) 3.67 (D) 4.67

2. When HNO_3 is converted into NH_3, the equivalent weight of HNO_3 will be:

(M = molecular weight of HNO_3)

(A) M/2 (B) M/1

(C) M/6 (D) M/8

3. In the ionic equation, $K^+ + BrO_3^- + 12H^+ + 10e^- \longrightarrow Br_2 + 6H_2O + 2K^+$, the equivalent weight of $KBrO_3$ will be:

(where M = molecular weight of $KBrO_3$)

(A) M/5 (B) M/2

(C) M/6 (D) M/4

4. If molecular weight of $KMnO_4$ is 'M', then its equivalent weight in acidic medium would be:

(A) M (B) M/2

(C) M/5 (D) M/4

5. In the conversion $NH_2OH \longrightarrow N_2O$, the equivalent weight of NH_2OH will be:

(M = molecular weight of NH_2OH)

(A) M/4 (B) M/2

(C) M/5 (D) M/1

6. In the reaction between SO_2 and O_3, the equivalent weight of ozone is:

(A) the same as its molecular weight

(B) half the molecular weight

(C) one-third of the molecular weight

(D) one-fourth of the molecular weight

7. The equivalent weight of phosphoric acid (H_3PO_4) in the reaction:

$NaOH + H_3PO_4 \longrightarrow NaH_2PO_4 + H_2O$

(A) 59 (B) 49

(C) 25 (D) 98

8. The equivalent weight of $MnSO_4$ is half its molecular weight when it is converted into

(A) Mn_2O_3 (B) MnO_4^-

(C) MnO_2 (D) MnO_4^{2-}

9. The equivalent weight of Mohr's salt, $FeSO_4.(NH_4)_2SO_4.6H_2O$ is equal to

(A) Its molecular weight

(B) Atomic weight

(C) half-its molecular weight

(D) one-third its molecular weight

10. 2g of a base whose eq. wt. is 40 reacts with 3g of an acid. The eq. wt. of the acid is-

(A) 40 (B) 60

(C) 10 (D) 80

11. How many millilitres of 0.1N H_2SO_4 solution will be required for complete reaction with a solution containing 0.125g of pure Na_2CO_3:

(A) 23.6 mL (B) 25.6 mL

(C) 26.3 mL (D) 32.6 mL

12. If 25 mL of a H_2SO_4 solution reacts completely with 1.06 g of pure Na_2CO_3, what is the normality of this acid solution:

(A) 1 N (B) 0.5 N

(C) 1.8 N (D) 0.8 N

13. How much water is to be added to dilute 10 mL of 10 N HCl to make it decinormal?

(A) 990 mL (B) 1010 mL

(C) 100 mL (D) 1000 mL

14. 250 ml of 0.1 N solution of $AgNO_3$ are added to 250 ml of a 0.1 N solution of NaCl. The concentration of nitrate ion in the resulting solution will be -

(A) 0.1N (B) 1.2 N

(C) 0.01 N (D) 0.05 N

15. 2g of NaOH and 4.9 g of H_2SO_4 were mixed and volume is made 1 litre. The normality of the resulting solution will be -
(A) 1N (B) 0.05 N
(C) 0.5 N (D) 0.1N

16. The volume of water to be added to 200 mL of semi normal HCl solution to make it decinormal is -
(A) 200 mL (B) 400 mL
(C) 600 mL (D) 800 mL

17. Normality of 1% H_2SO_4 solution is nearly -
(A) 2.5 (B) 0.1
(C) 0.2 (D) 1

18. Temporary hardness is due to bicarbonates of Mg^{2+} and Ca^{2+}. It is removed by addition of CaO as follows:
$Ca(HCO_3)_2 + CaO \longrightarrow 2CaCO_3 + H_2O$
Mass of CaO required to precipitate 2 g $CaCO_3$ is:
(A) 2 g (B) 0.56 g
(C) 0.28 g (D) 1.12 g

19. The mass of oxalic acid crystals ($H_2C_2O_4.2H_2O$) required to prepare 50 mL of a 0.2 N solution is:
(A) 4.5 g (B) 6.3 g
(C) 0.63 g (D) 0.45 g

20. 125 mL of 63% (w/v) $H_2C_2O_4.2H_2O$ solution is made to react with 125 mL of a 40% (w/v) NaOH solution.
The resulting solution is: (ignoring hydrolysis of ions)
(A) neutral (B) acidic
(C) strongly acidic (D) alkaline

21. If equal volumes of 0.1 M $KMnO_4$ and 0.1 M $K_2Cr_2O_7$ solutions are allowed to oxidise Fe^{2+} to Fe^{3+} in acidic medium, then Fe^{2+} oxidised will be:
(A) more by $KMnO_4$
(B) more by $K_2Cr_2O_7$
(C) equal in both cases
(D) cannot be determined

22. 1 mol H_2SO_4 can not neutralise:
(A) 2 mol of ammonia
(B) 1 mol of $Ba(OH)_2$
(C) 0.5 mol of $Ba(OH)_2$
(D) 2 mol of KOH

EXERCISE-2

1. In the balanced equation -
$aHgS + bHCl + cHNO_3 \longrightarrow dH_2HgCl_4 + eNO + fS + gH_2O$
The values of b, c, e, and g are respectively -
(A) 2, 2, 4, 12 (B) 6, 1, 1, 2
(C) 12, 2, 2, 4 (D) both (B) and (C)

2. In the reaction $FeSO_4 + x \longrightarrow Na_2SO_4 + Fe(OH)_2$, x is -
(A) Na_2SO_4 (B) H_2SO_4
(C) NaOH (D) none of these

3. In the reaction $CaO + H_2O \rightarrow Ca(OH)_2$ + heat energy, the possible state of $Ca(OH)_2$ is -
(A) liquid (B) aqueous
(C) gaseous (D) solid

4. Combustion is a process involving -
(A) oxidation
(B) reduction
(C) liberation of heat
(D) All (A), (B) & (C)

5. $aK_2Cr_2O_4 + bKCl + cH_2SO_4 \longrightarrow xCrO_2Cl_2 + y KHSO_4 + zH_2O$
The above equation balances when
(A) a = 2, b = 4, c = 6 and x = 2, y = 6, z = 3
(B) a = 4, b = 2, c = 6 and x = 6, y = 2, z = 3
(C) a = 6, b = 4, c = 6 and x = 6, y = 3, z = 2
(D) a = 1, b = 4, c = 6 and x = 2, y = 6, z = 3

6. In the reaction $Mg + Cl_2 \rightarrow MgCl_2$
Chlorine may be regarded as -
(A) an oxidizing agent
(B) a reducing agent
(C) a catalyst
(D) providing an inert medium

7. The antioxidant which is used to prevent rancidity in foods is
(A) butylated hydroxyl anisole
(B) sodium hydroxide
(C) sodium carbonate
(D) methylated hydroxyl anisole

8. The colour of the chemical that changes when it is left open in sunlight is -
(A) $FeSO_4(s)$ (B) $Pb(NO_3)_2(s)$
(C) $AgCl(s)$ (D) $BaCl_2(aq)$

9. Sun produces heat and light by -
(A) combustion
(B) inflammation
(C) nuclear reactions
(D) photochemical reactions

10. Heating of calcium carbonate to give calcium oxide and carbon dioxide is -
(A) An oxidation process
(B) A reduction process
(C) Disproportionation
(D) Decomposition

11. Which of the following statement(s) is/are correct?
(A) Many compounds can be used as oxidizing as well as reducing agent
(B) We need equal moles and equal volumes of strong monobasic acid and strong monoacidic base to completely neutralize themselves
(C) The quantity of a catalyst does not change at the end of a chemical reaction
(D) All of these are correct

12. The elements undergoing reduction and oxidation during the reaction,
$(NH_4)_2Cr_2O_7 \longrightarrow N_2 + Cr_2O_3 + 4H_2O$
are respectively:
(A) N and Cr (B) Cr and N
(C) N and O (D) Cr and O

13. The decomposition of $KClO_3$ to KCl and O_2 on heating is an example of:
(A) Intermolecular redox change
(B) Intramolecular redox change
(C) Disproportionation or auto redox change
(D) None of the above

14. Which of the following can act as oxidant?
(A) H_2O_2 (B) H_2S
(C) NH_3 (D) None of these

15. Conversion of $PbSO_4$ to PbS is -
(A) reduction of S (B) oxidation of S
(C) dissociation (D) none of these

16. Which of the following is redox reaction?
(A) $NaOH + HCl \longrightarrow NaCl + H_2O$
(B) $AgNO_3 + KI \longrightarrow AgI + KNO_3$
(C) $BaO_2 + H_2SO_4 \longrightarrow BaSO_4 + H_2O_2$
(D) $SnCl_2 + HgCl_2 \longrightarrow SnCl_4 + Hg$

17. In a conjugate pair of reductant and oxidant, the reductant has:
(A) Lower ox. no. (B) Higher ox. no.
(C) Same ox. no. (D) Either of these

18. Oxidation is process which involves:
(A) De-electronation
(B) Electronation
(C) Addition of hydrogen
(D) Addition of metal

19. Oxidants are substances which:
(A) Show a decrease in their oxidation number during a change
(B) Gain electrons during a change
(C) Oxidize others and reduce themselves
(D) All of the above

20. If an iron rod is dipped in $CuSO_4$ solution:
(A) Blue colour of the solution turns red
(B) Brown layer is deposited on iron rod
(C) No change occurs in the colour of the solution
(D) None of the above

21. Which of the following is a redox reaction?
(A) $CaCO_3 \longrightarrow CaO + CO_2$
(B) $H_2 + Cl_2 \longrightarrow 2HCl$
(C) $CaO + 2HCl \longrightarrow CaCl_2 + H_2O$
(D) $NaOH + HCl \longrightarrow NaCl + H_2O$

22. Which of the following statement about the following reaction is correct?

$ZnO + CO \longrightarrow Zn + CO_2$

(A) ZnO is being oxidized
(B) CO is being reduced
(C) CO_2 is being oxidized
(D) ZnO is being reduced

23. Calorific value of a fuel may be defined as -

(A) The amount of heat produced when 1000 kg of a fuel is completely burnt.
(B) The amount of heat produced when 1g of fuel is incompletely burnt.
(C) The amount of heat produced when 10 g of a fuel is completely burnt.
(D) The amount of heat produced in kilojoules when unit mass of a fuel is completely burnt.

24. Which of the following fuels has the highest calorific value?

(A) Petrol (B) Hydrogen
(C) LPG (D) Natural gas

25. In which zone of a candle flame does complete combustion take place?

(A) Inner (B) Outer
(C) Middle (D) All three zones

26. The zone of no combustion is -

(A) the middle zone
(B) outermost zone
(C) the base of the flame
(D) innermost zone.

27. The colour of the flame in luminous zone is -

(A) blue (B) red
(C) yellow (D) black

28. The non-combustible substance among the following is -

(A) coal (B) glass
(C) wood (D) paper

29. Combustion is a -

(A) physical process
(B) chemical process
(C) biological process
(D) mechanical process

30. A circular blackish ring on the glass plate kept over the candle flame indicates-

(A) complete combustion
(B) dark zone of the flame
(C) non-luminous zone
(D) luminous zone

31. The oxidation number of sulphur in S_8, S_2F_2 and H_2S respectively are:

(A) 0, + 1 and – 2 (B) + 2, + 1 and – 2
(C) 0, + 1 and + 2 (D) – 2, + 1 and – 2

32. Which of the following compounds have sulphur atom in its maximum oxidation state:

$X = H_2SO_4$; $Y = H_2SO_5$; $Z = H_2S_2O_8$

(A) Z only (B) Y and Z only
(C) Y only (D) X, Y and Z

33. Manganese achieves its maximum oxidation state in which of these compounds:

(A) MnO_2 (B) Mn_3O_4
(C) $KMnO_4$ (D) K_2MnO_4

34. Oxidation number of underlined elements in N_2O_5, SO_3^{2-}, NH^+_4 are:

(A) +5, +2, –3 (B) +6, –2, +3
(C) +6, +2, –3 (D) +5, +4, –3

35. Which of the following changes does not involve either oxidation or reduction:

(A) $VO^{2+} \rightarrow V_2O_3$
(B) $Na \rightarrow Na^+$
(C) $Zn^{+2} \rightarrow Zn$
(D) $CrO_4^{-2} \rightarrow Cr_2O_7^{-2}$

36. Which of the following is not a redox reaction:

(A) $Mg + N_2 \longrightarrow Mg_3N_2$
(B) $MnO_4^- + C_2O_4^{2-} \longrightarrow Mn^{2+} + CO_2$
(C) $CuSO_4 + KI \longrightarrow 2CuI + I_2 + K_2SO_4$
(D) $AgCl + NH_3 \longrightarrow [Ag(NH_3)_2]Cl$

37. Vapour density of a metal chloride is 66. Its oxide contains 53% metal. The atomic weight of metal is -
(A) 21 (B) 54
(C) 26.74 (D) 2.086

38. Equivalent mass of $KMnO_4$, when it is converted to $MnSO_4$ is -
(A) M/5 (B) M/3
(C) M/6 (D) M/2

39. In a compound A_xB_y -
(A) Mole of A = Mole of B = mole of A_xB_y
(B) Eq. of A = Eq. of B = Eq. of A_xB_y
(C) X × mole of A = y × mole of B = (x + y) × mole of A_xB_y
(D) X × mole of A = y × mole of B

40. A certain weight of pure $CaCO_3$ is made to react completely with 200 mL of a HCl solution to give 224 mL of CO_2 gas at STP. The normality of the HCl solution is:
(A) 0.05N (B) 0.1 N
(C) 1.0 N (D) 0.2 N

41. 0.2 g of a sample of H_2O_2 required 10 mL of 1N $KMnO_4$ in a titration in the presence of H_2SO_4. Purity of H_2O_2 is-
(A) 25% (B) 85%
(C) 65% (D) 95%

42. 150 ml of 10N HCl is required to react completely with 1.0 g of a sample of limestone. The percentage purity of calcium carbonate is -
(A) 75% (B) 50%
(C) 80% (D) 90%

43. An oxide of metal have 20% oxygen, the eq. wt. of metal oxide is -
(A) 32 (B) 40
(C) 48 (D) 52

44. For the redox reaction $MnO_4^- + C_2O_4^{2-} + H^+ \longrightarrow Mn^{2+} + CO_2 + H_2O$, the correct stoichiometric coefficients of MnO_4^-, $C_2O_4^{2-}$ and H^+ are respectively:
(A) 2, 5, 16 (B) 16, 5, 2
(C) 5, 16, 2 (D) 2, 16, 5

45. In the half reaction $2ClO_3^- \rightarrow Cl_2$
(A) 5 electrons are gained
(B) 5 electrons are liberated
(C) 10 electrons are gained
(D) 10 electrons are liberated

46. 1g of a metal carbonate neutralises completely 200 mL of 0.1N HCl. The equivalent weight of metal carbonate is
(A) 25 (B) 50
(C) 100 (D) 75

47. 100 mL of 0.5 N NaOH were added to 20 ml of 1N HCl and 10 mL of 3 N H_2SO_4. The solution is -
(A) acidic (B) basic
(C) neutral (D) none of these

48. 1M solution of H_2SO_4 is diluted from 1 litre to 5 litres, the normality of the resulting solution will be -
(A) 0.2 N (B) 0.1 N
(C) 0.4 N (D) 0.5 N

49. 0.7 g of $Na_2CO_3.xH_2O$ is dissolved in 100 mL. 20 mL of which required to neutralize 19.8 mL of 0.1 N HCl. The value of x is -
(A) 4 (B) 3
(C) 2 (D) 1

50. 0.45 g of an acid of molecular weight 90 was neutralised by 20 mL of 0.5 N caustic potash. The basicity of the acid is -
(A) 1 (B) 2
(C) 3 (D) 4

51. 1 litre of 18 molar H_2SO_4 has been diluted to 100 litres. The normality of the resulting solution is -
(A) 0.09 N (B) 0.18
(C) 1800 N (D) 0.36

52. 50 ml of $\frac{N}{10}$ HCl is treated with 70 ml $\frac{N}{10}$ NaOH. Resultant solution is neutralized by 100 ml of sulphuric acid. The normality of H_2SO_4 -
(A) N/50 (B) N/25
(C) N/30 (D) N/10

53. 200 mL of $\frac{N}{10}$ HCl were added to 1 g calcium carbonate, what would remain after the reaction?
(A) $CaCO_3$ (B) HCl
(C) Neither of the two (D) Part of both

54. An element A in a compound ABD has oxidation number –n. It is oxidised by $Cr_2O_7^{2-}$ in acid medium. In the experiment, 1.68×10^{-3} moles of $K_2Cr_2O_7$ were used for 3.36×10^{-3} moles of ABD. The new oxidation number of A after oxidation is:
(A) 3 (B) 3 – n
(C) n – 3 (D) +n

55. The eq. wt. of an element is 4. Its chloride has a VD 59.25. Then the valency of the element is.
(A) 4 (B) 3
(C) 2 (D) 1

56. If 250 mL of a solution contains 24.5 g H_2SO_4 the molarity and normality respectively are -
(A) 1 M, 2 N (B) 1M, 0.5 N
(C) 0.5 M, 1N (D) 2M, 1N

57. Which of the following has the highest normality?
(A) 1 M H_2SO_4 (B) 1 M H_3PO_3
(C) 1 M H_3PO_4 (D) 1 M HNO_3

58. What mass of $Na_2S_2O_3.5H_2O$ is needed to make 500 cm^3 of 0.200 N solution for the reactions
$2S_2O_3^{2-} + I_2 \rightarrow S_4O_6^{2-} + 2I^-$
(A) 4.8 gm (B) 14.8 gm
(C) 24.8 gm (D) 16.8 gm

59. What volume of 3 molar HNO_3 is needed to oxidise 8 g of Fe^{2+} to Fe^{3+}, HNO_3 gets converted to NO:
(A) 8 mL (B) 16 mL
(C) 32 mL (D) 64 mL

60. The reaction,
$2K_2MnO_4 + Cl_2 \rightarrow 2KMnO_4 + 2KCl$ is an example of -
(A) Oxidation (B) Reduction
(C) Redox (D) Chlorination

61. In the reaction, $3Br_2 + 6CO_3^{2-} + 3H_2O \rightarrow 5Br^- + BrO_3^- + 6HCO_3^-$
(A) Bromine is oxidised and carbonate is reduced
(B) Bromine is oxidised as well as reduced
(C) Bromine is reduced and water is oxidised
(D) Br_2 is neither oxidised nor reduced

62. The charge on cobalt in $[Co(CN)_6]^{3-}$ is-
(A) – 6 (B) – 3
(C) + 3 (D) + 6

63. In the chemical reaction, $K_2Cr_2O_7 + X\ H_2SO_4 + Y\ SO_2 \rightarrow K_2SO_4 + Cr_2(SO_4)_3 + ZH_2O$ X, Y and Z are -
(A) 1, 3, 1 (B) 4, 1, 4
(C) 3, 2, 3 (D) 2, 1, 2

64. Select the incorrect statement from the following option.
(A) Water which does not form lather with soap and forms white scum is called hard water
(B) Hard water contains dissolved calcium and magnesium salts in it
(C) In hard water, cleansing quality of soap is depressed
(D) Due to the presence of dissolved hardness-producing salts, the boiling point of water is depressed

65. Select the incorrect statement from the following option.
(A) Permanent hardness is due to dissolved chlorides and sulphates of calcium and magnesium
(B) It can be removed by mere boiling of water
(C) It is also known as non-alkaline hardness
(D) The difference between the total hardness and the alkaline hardness gives the non-alkaline hardness

66. The volume strength of 1.5 N H_2O_2 is
(A) 4.8 (B) 8.4
(C) 3.0 (D) 8.0

NSEJS PYQ

1. Oxygen exhibits (–1) oxidation state in **[NSEJS-2011]**

(A) OF_2 (B) H_2O

(C) H_2O_2 (D) HClO

2. The oxidation number of chlorine in $CaOCl_2$ is- **[NSEJS-2013]**

(A) 0 (B) –1

(C) +1 (D) +3

3. Oxidation number and co-ordinaiton number of Pt in cisplatin $PtCl_2(NH_3)_2$ are respectively **[NSEJS-2014]**

(A) +4 and 2

(B) +2 and 4

(C) 0 and 4

(D) +2 and 6

4. The method that can not be used for removing permanent hardness of water is: **[NSEJS-2014]**

(A) Adding sodium carbonate

(B) Distillation

(C) Adding Caustic soda

(D) Boiling

5. Amongst the sixteen ✓ and × marks, how many cases are incorrect? **[NSEJS-2015]**

Strip of Metal	0.5M $MgSO_4$	0.5M $ZnSO_4$	0.5M $PbSO_4$	0.5M $FeSO_4$
(i) Mg	×	✓	×	✓
(ii) Zn	×	×	×	×
(iii) Fe	✓	✓	✓	×
(iv) Pb	✓	✓	✓	✓

✓ means displacement reaction occurs

× means no displacement reaction occurs

(A) 7 (B) 11

(C) 10 (D) 9

6. What is the ratio of number of electrons gained by acidified $KMnO_4$ and acidified $K_2Cr_2O_7$ in the reaction given below **[NSEJS-2015]**

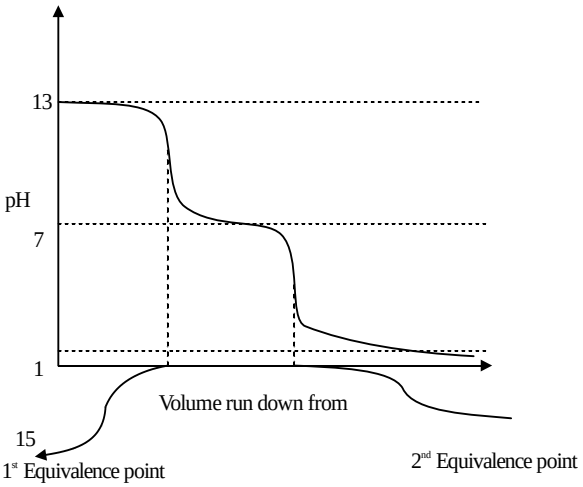
$$KMnO_4 \xrightarrow{H^+} Mn^{2+}$$

$$K_2Cr_2O_7 \xrightarrow{H^+} Cr^{3r}$$

(A) 5 : 6 (B) 6 : 5

(C) 3 : 5 (D) 5 : 3

7. Given below is the pH vs volume curve for titration of Na_2CO_3 (in conical flask) with HCl. **[NSEJS-2015]**



Choose the correct statement based on the above graph

(A) 1st equivalence point represents $NaHCO_3$ getting protonated to H_2CO_3.

(B) 2nd equivalence point represents $NaHCO_3$ getting protonated to H_2CO_3.

(C) 1st equivalence point represents deprotonation of HCl by $NaHCO_3$ alone.

(D) 2nd equivalence point represents deprotonation of HCl by Na_2CO_3 alone.

8. Rust is a mixture of **[NSEJS-2015]**

(A) $FeO + Fe(OH)_2$ (B) $FeO + Fe(OH)_3$

(C) $Fe_2O_3 + Fe(OH)_3$ (D) $Fe_3O_4 + Fe(OH)_3$

9. The oxidation number of sulphur in sodium thiosulphate ($Na_2S_2O_3$) is **[NSEJS-2015]**

(A) +1 (B) +2

(C) +3 (D) +4

10. The Equivalent weight of $MnSO_4$ is half its molecular weight when it is converted to:

[NSEJS-2015]

(A) Mn_2O_3 (B) MnO_4

(C) MnO_2 (D) MnO_4^{-2}

11. Identify the overall change in the following set of reactions: **[NSEJS-2016]**

1. Carbon dioxide $\rightarrow$ carbonic acid (H_2CO_3)
2. Ethanol (alcohol) $\rightarrow$ Ethanal (aldehyde)
3. Ethanal (aldehyde) $\rightarrow$ Ethanol (alcohol)
4. Sulphuric acid $\rightarrow$ Sulphur trioxide (SO_3)

Choose the correct option which best describes these conversions

(A) oxidation, oxidation, reduction, reduction

(B) hydration, oxidation, reduction, dehydration

(C) reduction, dehydration, hydration, oxidation

(D) reduction, reduction, oxidation, oxidation

12. Sati was studying neutralization reaction. She accidentally dropped 'x' grams of a sodium carbonate monohydrate into a 100 mL solution of HCl whose concentration was 0.25M. 10mL from this final solution was taken and titrated against 0.05M NaOH solution to yield a titre value of 20mL. What is the value of 'x'.

[NSEJS- 2016]

(A) 186mg (B) 93mg

(C) 1860mg (D) 930mg

13. Which is incorrect **[NSEJS--2017]**

(I) $Zn + CuSO_4$ (aq) $\rightarrow$ Reaction occurs

(II) $Zn + Al_2(SO_4)_3$(aq) $\rightarrow$ Reaction does not occur

(III) $Zn + AgNO_3$ (aq) $\rightarrow$ Reaction does not occur

(IV) $Zn + PbNO_3$ (aq) $\rightarrow$ Reaction occurs

(A) I (B) II

(C) III (D) IV

14. When 1 gram of a mixture of aluminium and zinc was treated with HCl, a gas was liberated. At the end of the reaction, the volume of the liberated gas was found to be 524 cm^3, under STP conditions. The individual weights of aluminium and zinc in the mixture, respectively, are: **[NSEJS-2017]**

(A) 0.2 g and 0.8 g

(B) 0.8 g and 0.2 g

(C) 0.5 g and 0.5 g

(D) 0.322g and 0.678 g

15. Every major city in India has a pollution control board to monitor air and water pollution. The following data is from three different localities in Bangalore city from the year 2015. **[NSEJS-2017]**

Locality	**Annual average of SO_2 in the air (volume/volume)**
P	16.3 mL/m^3
Y	16.3 ppb (m^3/ m^3)
Z	16.3 ppm (m^3/ m^3)

ppb stands for parts per billion and ppm stands for parts per million. These are different units to express. concentration. They are very similar to percentage (which is actually parts per hundred). Based on the above data, which place will you choose to live in?

(A) All localities are equally polluted, so I have no preference.

(B) P is the more polluted tha Y and Z, hence I will live in either Y or Z.

(C) Locality Y is least polluted, hence I will live in Y.

(D) Z and Y are more polluted than P, hence I will live in P.

16. $2KBrO_3 + 12H^+ + 10e^- \rightarrow Br_2 + 6H_2O + 2K^+$
From above reaction the equivalent weight of $KBrO_3$ can be calculated as (M is molecular weight of $KBrO_3$) **[NSEJS-2018]**
(A) M/5 (B) M/10
(C) M/12 (D) M/2

17. Shaila took about 10 cm^3 of a diluted Potassium hydrogen carbonate solution in a test tube. To this solution she added few drops of universal indicator. The colour of the solution turned: **[NSEJS-2018]**
(A) orange (B) green
(C) blue (D) yellow

18. What will happen if a copper piece is dipped in aqueous solution of silver nitrate for quite some time? **[NSEJS- 2018]**
(i) Solution will remain colourless
(ii) Solution will turn blue
(iii) Silver will deposit on the copper piece
(iv) Bubbles of brown gas will be formed around copper piece
(A) i and iv (B) ii and iv
(C) ii and iii (D) iii and iv

19. A solution of pure aluminium sulphate containing 0.170 g of aluminium ions is treated with excess of barium hydroxide solution. Total weight of the precipitate will be: **[NSEJS-2018]**
(A) 0.5 g (B) 2.7 g
(C) 1.7 g (D) 0.54 g

20. A strip of iron with mass 15.5 g is placed in a solution containing 21.0 g copper sulphate. After some time the reaction stops. Iron strip was found to have mass 8.5 g. The mass of copper formed was found to be 8.60 g. Find the mass of ferrous sulphate formed in this reaction. **[NSEJS-2018]**
(A) 19.40 g (B) 18.40 g
(C) 17.40 g (D) 16.40 g

21. A white solid known to be a compound of sodium, given rise to water vapour and a colourless gas on heating. The residual white powder is dissolved in water and when the solution is added to alum solution, a white gelatinous precipitate is obtained. The original solid was: **[NSEJS-2018]**
(A) Sodium carbonate
(B) Sodium bicarbonate
(C) Sodium hydroxide
(D) Sodium nitrate

22. A teacher wanted to give acid base titration to her students. For that she prepared (i) HCl solution by dissolving 73 g of hydrochloric acid in one litre of water and (ii) sodium hydroxide solution by dissolving 0.46 g of sodium metal in one litre of water. Find the volume of the hydrochloric acid solution required for complete neutralisation of sodium hydroxide solution. **[NSEJS-2018]**
(Cl= 35.5; Na = 23.0; O = 16.0)
(A)20 mL (B) 10 mL
(C) 46 mL (D) 5 mL

23. Harsha was trying to neutralize phosphoric acid using various bases. Those available were caustic soda, lime water and hydrated alumina. If Harsha took 1 equivalent of phosphoric acid each time, what will be the ratio for moles of each of the above bases required for complete neutralization? **[NSEJS-2018]**
(A) 1: 1: 1 (B) 1: 0.5: 0.33
(C) 1: 2: 3 (D) 1: 0.33: 0.5

24. What is the ratio of reducing agent to oxidizing agent, if the following reaction is correctly balanced?
$NH_3 + O_2 \rightarrow NO + H_2O$ **[NSEJS-2019]**
(A) 4:5 (B) 5:4
(C) 5:3 (D) 3:5

25. Sonu has N/2 HCl solution and Monu has N/10 HCl solution. They are asked to prepare 2 litres of N/5 HCl solution. What volume of two solutions be mixed? **[NSEJS-2019]**

(A) (0.5 + 1.5) litre (B) (1.0 + 1.0) litre

(C) (0.3 +1.7) litre (D) (0.2 +1.8) litre

26. A solution (P) was prepared by dissolving 6.3 g of oxalic acid in 100 ml water. 25 ml of this solution was taken and was further diluted to 250 ml to prepare solution (Q). What weight of NaOH in ppm will be required to neutralize 10 ml of solution (Q)? **[NSEJS-2019]**

(A) 40 ppm (B) 10 ppm

(C) 20 ppm (D) 80 ppm

27. $2KBrO_3 + 12H^+ + 10e^- \rightarrow Br_2 + 6H_2O + 2K^+$ From above reaction the equivalent weight of $KBrO_3$ can be calculated as (M is molecular weight of $KBrO_3$) **[NSEJS-2019]**

(A) M/5 (B) M/10

(C) M/12 (D) M/2

28. In an experiment with 100 mL 0.1 M solution of Copper Chloride, by mistake 5 gms of a mixture containing equal weights of Tin, Silver, Lead and Calcium, was added. Finally after some time the solution gets completely decolorized. This is mainly due to:

[NSEJS-2021]

(A) Silver reacts with Copper Chloride

(B) Calcium reacts with Copper Chloride

(C) All the metals react with Copper Chloride

(D) Only Lead reacts with Copper Chloride forming white precipitate of lead chloride

29. Equal lengths of magnesium ribbons are taken in four test tubes A, B, C and D. In test tube A, 1M acetic acid is added; in test tube B, 1M HCI is added; in test tube C, 1M HNO_3 is added; and in test tube D, 1M NaOH is added. The observed results will be: **[NSEJS-2021]**

(A) The fizzing occurs more vigorously in A

(B) The fizzing occurs more vigorously in B

(C) The fizzing occurs more vigorously in C

(D) The fizzing occurs more vigorously in D

30. Sum of oxidation states of all the carbon atoms in toluene molecule is: **[NSEJS-2022]**

(A) –1 (B) $-\frac{7}{8}$

(C) $-\frac{8}{7}$ (D) –8

31. Oxidation state of oxygen in O_2PtF_6 is:

[NSEJS-2022]

(A) $-\frac{1}{2}$ (B) –2

(C) $\frac{1}{2}$ (D) –1

32. A mass 0.75 g of the mixture of Na_2CO_3 and K_2CO_3 is completely neutralized by 50 mL 0.25 N HCl. The percentage of Na_2CO_3 in the mixture is: **[NSEJS-2023]**

(A) 50.6 (B) 49.4

(C) 50 (D) data insufficient

33. The reaction $KI + I_2 \rightarrow KI_3$ involves

[NSEJS-2023]

(A) oxidation

(B) reduction

(C) complex formation

(D) neutralization

ADVANCED & OLYMPIAD CHALLENGER QUESTIONS

1. $Ag(s) + NO_3(aq) + H^+(aq) \rightarrow Ag^+(aq) + NO(g) + H_2O(1)$

When the oxidation-reduction equation above is balanced, what is the coefficient for $H^+(aq)$?

(A) 1 (B) 2

(C) 3 (D) 4

2. Which equation represents an oxidation-reduction reaction?

(A) $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

(B) $H_2SO_4 + Ca(OH)_2 \rightarrow CaSO_4 + 2H_2O$

(C) $MgCrO_4 + BaCl_2 \rightarrow MgCl_2 + BaCrO_4$

(D) $Zn(NO_3)_2 + Na_2CO_3 \rightarrow 2NaNO_3 + ZnCO_3$

3. In the chemical reaction

$3\ Br_2 + 6CO_3^{-2} + 3\ H_2O \rightarrow BrO_3^- + 5Br^- + 6HCO_3^-$

(A) bromine is reduced and water is oxidized

(B) bromine is oxidized and carbonate is reduced

(C) bromine is both oxidized and reduced

(D) bromine is neither oxidized nor reduced

4. A solution of H_2SO_4 is 80% by weight, having specific gravity 1.73. Its normality is:

(A) 18.0 (B) 28.2

(C) 1.0 (D) 10.0

5. In the reaction

(i) $BaO_{2\,(s)} + O_{3\,(g)} \rightarrow BaO_{(s)} + 2O_{2\,(g)}$

(ii) $ZnS_{(s)} + 4O_{3\,(g)} \rightarrow ZnSO_{4\,(s)} + 4O_{2\,(g)}$

(A) O_3 is reduced both in (i) & (ii)

(B) O_3 is oxidized both in (i) & (ii)

(C) O_3 is oxidized in (i) & reduced in (ii)

(D) O_3 is reduced in (i) & oxidized in (ii)

6. When pressure is applied to equilibrium system,

$Ice_{(s)} \neq H_2O_{(l)}$

which of the following will happen?

(A) More ice will be formed

(B) Water will evaporate

(C) More water will be formed

(D) There will be no change

7. Which of the following is an oxidation-reduction reaction?

(A) $H_3O^+(aq) + CO_3^{2-}(aq) \rightarrow HCO_3^-(aq) + H_2O$

(B) $HNO_3(aq) + NH_3(aq) \rightarrow NH_4^+(aq) + NO_3^-(aq)$

(C) $Mg(s) + F_2(g) \rightarrow MgF_2(s)$

(D) $Pb(NO_3)_2(aq) + 2\ NaCl(aq) \rightarrow PbCl_2(s) + 2NaNO_3(aq)$

8. Rajesh was performing the thermometric titration with caustic potash and oil of vitriol, under the same set of conditions. He dissolved 19.6 g of oil of vitriol in $1dm^3$ of water and 5.6g of caustic potash in 1 dm^3 of water. Find the volume of oil of vitriol taken initially for thermometric titration. The observation table of his experiment is as follows.

Obs. No.	Volume of KOH added/cm^3	Temperature °C
1	20	27
2	30	27.4
3	40	27.5
4	50	27.8
5	60	27.6
6	70	27.2
7	80	26.9

(A) 6.25 cm^3 (B) 50 cm^3

(C) 12.5 cm^3 (D) 25 cm^3

9. Titration is carried out for 3 moles of a mixture of ferrous and ferric sulphate, which required 100 mL of 2 M potassium permanganate in acidic medium. Hence the mole fraction of ferrous sulphate in the mixture is _____.

(A) 1/3 (B) 2/3

(C) 2/5 (D) 3/5

10. In a chemistry laboratory, a student found a bottle labeled 'Acid'. As it was a solid, she was curious to find out what it is. She weighed 0.42g of this and made a solution of it and titrated it with 0.17M NaOH solution. The volume of NaOH required to obtain the end point was 33.8 mL. If the molecular formula of the acid is $C_6H_{10}O_4$, find out the number of protons per acid molecule that take part in the reaction and the amount of acid required to neutralize 1mole of the alkali.

(A) 1 proton and 73 g
(B) 2 protons and 146 g
(C) 1 proton and 46 g
(D) 2 protons and 73 g

11. Iron present in spinach can be estimated by titrating it with potassium permanganate. Small amounts of spinach leaves are weighed and dissolved in acid to extract the iron in solution. The solution is then titrated and the following reaction takes place during this titration.

$$__Fe^{2+} + __ MnO_4^- + __H^+ \rightarrow __Mn^{2+} + __Fe^{3+} + __H_2O$$

When properly balanced with the simplest set of whole number coefficients, the sum of the coefficients in the balanced equation is

(A) 16 (B) 18
(C) 22 (D) 24

12. A disproportionation reaction occurs with a simultaneous oxidation and reduction of the same species in the reaction. Which of the following is NOT a disproportionation reaction?

(A) $2NO_2 + H_2O \rightarrow HNO_3 + HNO_2$
(B) $3S + 2H_2O \rightarrow SO_2 + 2H_2S$
(C) $NH_4NO_3 \rightarrow N_2O + 2H_2O$
(D) $3Cl_2 + 6OH^- \rightarrow 5Cl^- + ClO_3^- + 3H_2O$

13. Upon fully dissolving 2.0 g of a metal in sulfuric acid, 6.8 g of the metal sulfate is formed. The equivalent weight of the metal is:

(A) 13.6 g (B) 20.0 g
(C) 4.0 g (D) 10.0 g

14. Complete reaction of 2.0 g of calcium (at. wt. = 40) with excess HCl produces 1.125 L of H_2 gas. Complete reaction of the same quantity of another metal "M" with excess HCl produces 1.85 L of H_2 gas under identical conditions. The equivalent weight of "M" is closest to

(A) 23 (B) 9
(C) 7 (D) 12

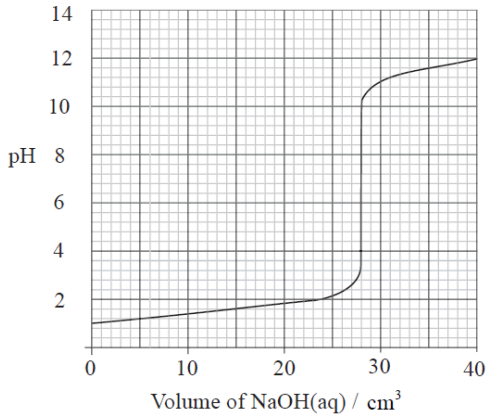
15. The number of electrons required to reduce chromium completely in $Cr_2O_7^{2-}$ to Cr^{3+} in acidic medium, is

(A) 5 (B) 3
(C) 6 (D) 2

16. $KMnO_4$ reacts with H_2O_2 in an acidic medium. The number of moles of oxygen produced per mole of $KMnO_4$ is

(A) 2.5 (B) 5
(C) 1.25 (D) 2

17. A titration was carried out to determine the concentration of 25.0 cm^3 of an aqueous solution of nitric acid. The pH value of the liquid in the flask was measured as 0.100 mol dm^{-3} of aqueous sodium hydroxide was added. The results are shown on the graph below.



(i) Use the graph to determine the volume of 0.100 mol dm^{-3} aqueous sodium hydroxide solution needed to exactly neutralize the nitric acid.
(ii) Determine the pH value when the value of $[H^+]$ has decreased to 1×10^{-3} mol dm^{-3}
(iii) Use the graph to determine the value of $[H^+]$ of the nitric acid solution.

18. A mixture of $H_2C_2O_4$ (oxalic acid) and $NaHC_2O_4$ weighing 2.02 g was dissolved in water and this solution was made upto one liter. Two titrations were carried out and the result obtained: 10 ml of the solution required 3 ml of 0.1NaOH for complete neutralization. Calculate the amount of $H_2C_2O_4$ and $NaHC_2O_4$ in the mixture?

19. Aruna was studying the interaction between various chemicals. She found that when she mixed potassium bromate ($KBrO_3$) with potassium bromide (KBr) and acidified the solution with H_2SO_4, vapours of Br_2 were evolved. On addition of KI to the reaction mixture, a deep brown colour was obtained due to formation of iodine, and this could be titrated with sodium thiosulphate ($Na_2S_2O_3$) using starch indicator.

i. Write down the balanced reaction occurring at each stage.

ii. In what molar proportion should $KBrO_3$ and KBr be mixed?

iii. Starting with 10 ml of 0.01M $KBrO_3$ and using excess of KBr, H_2SO_4 & KI, what volume of 0.05 M $Na_2S_2O_3$ will be required?

20. Acid base reactions are extremely common in nature and therefore it is of utmost interest to a chemist. A chemist gets a sample of drain cleaner and wants to find out the exact amount of sodium hydroxide present in it. He uses 23mL of 0.9M Phosphoric acid to completely neutralize the base.

(i) Write the balanced chemical equation for the reaction.

(ii) How many grams of Sodium hydroxide did the chemist find in the sample?

(iii) A chemist needs to find the molarity of 10% w/w HCl. What is the molarity of the solution if the density of the solution is 1.047 g/cm3 and molecular weight of HCl is 36.5g/mole.

(iv) There are many acid base reactions that take place in the body. If the acid content in the stomach is increased, antacids are used to neutralize it. A particular sample of antacid tablet contains sodium bicarbonate. When ingested, it reacts with the gastric juice (hydrochloric acid) in the stomach to give off carbon dioxide gas. When a 1.2 g tablet reacted with 40.00 mL of hydrochloric acid (density: 1.140 g/mL), carbon dioxide gas was evolved and the resulting solution weighed 46.7 g. Calculate the volume of carbon dioxide gas released at STP if its density is 1.98 g/L.

21. For quantitative analysis 3 g of mixture of sodium carbonate, sodium bicarbonate and sodium chloride was supplied to students. They found out on gentle heating, the mixture liberates 56 mL of CO_2 at NTP and another 3g of the same mixture requires 30.5mL of 1N hydrochloric acid for complete neutralization. Calculate the percentage of sodium chloride. [2.0]

22. A sample of water which gives lather with soap with difficulty is known as hard water, while a sample of water which gives lather with soap easily is known as soft water.

Hardness of water is due to the presence of bicarbonates, sulphates and chlorides of calcium and magnesium. Hardness of water is of two types, temporary and permanent hardness. When hardness of water is due to the presence of bicarbonates of magnesium and calcium, it is called temporary hardness. When hardness of water is due to the presence of sulphates and chlorides of magnesium and calcium, it is called permanent hardness.

The amount of hardness causing substances in a certain volume of water measures the extent of hardness or degree of hardness. Hardness of water is always calculated in terms of calcium carbonate although it is never responsible for causing hardness of water because of its insoluble nature.

The reason for choosing calcium carbonate as the standard for calculating hardness of water is the ease in calculation as its molecular weight is exactly 100.

Degree of hardness is usually expressed as parts per million (ppm) and thus may be defined as the number of parts by weight of calcium carbonate (equivalent to calcium and magnesium salt) present in a million (10^6) parts by weight of water.

1ppm = 1 part of $CaCO_3$ in (10^6) parts of water.

(I) Vishal has two samples of hard water, one contains 2mg of calcium sulphate and 0.5mg of magnesium chloride per litre of water and another contains 3mg magnesium sulphate per kg of water. Calculate the degree of hardness of both the samples.

(II) Permanent hardness of water can be removed by adding washing soda, if both temporary and permanent hardness are present together, then water is softened by addition of caustic soda. Give balanced chemical equations for softening of hard water.

23. Acid rain is a term referring to rain having a pH lower than that of natural rain. Historic monuments built with various materials such as iron coated with layers of $CaCO_3$ and Na_2SO_4 can get damaged by acid rain. Acid rain can lead to flaking of this coat. One such sample of coating was brought to the lab to be analysed. The weight of sample was 0.626 g.

The analyst added the sample to aqueous oxalic acid and completely precipitated the calcium as calcium oxalate (CaC_2O_4). The calcium oxalate precipitate obtained was then dissolved in sulphuric acid and the resulting oxalic acid ($H_2C_2O_4$) formed was titrated with a standard $KMnO_4$ solution. The titration of the oxalic acid required 17.8 mL of 0.l M $KMnO_4$ solution.

(A) Balance the equation for the titration reaction between $KMnO_4$ and $H_2C_2O_4$.

_ $KMnO_4$ + _ $H_2C_2O_4$ + _ H_2SO_4 $\rightarrow$ _ K_2SO_4 + $MnSO_4$ + _ H_2O + _ CO_2

(B) Identify the oxidizing agent and the reducing agent in the reaction.

(i) _____ is an oxidizing agent.

(ii)_____ is a reducing agent.

(C) Calculate the number of moles of oxalic acid reacted with the $KMnO_4$.

Show your calculations in the box.

Answer: _____ moles of oxalic acid

(D) Calculate the mass (in g) of $CaCO_3$ in the original sample.

Show your calculations in the box.

Answer: _____ g of $CaCO_3$

(E) Find the percent (%) of Na_2SO_4 present in the original ample.

Show your calculations in the box.

Answer: _____

Answer Key

EXERCISE-1

General Introduction of Chemistry & Characteristics of Chemical Reactions

1. (D) **2.** (D) **3.** (A) **4.** (B)
5. (C) **6.** (C) 7. (C) **8.** (A)
9. (B) **10.** (A)

Chemical Equations & Balancing

1. (C) **2.** (B) **3.** (C) **4.** (C)
5. (B) **6.** (A)

Redox Reactions & Oxidation Number

1. (B) **2.** (B) **3.** (D) **4.** (A)
5. (A) **6.** (C) **7.** (B) **8.** (D)
9. (C) **10.** (D) **11.** (D) **12.** (C)

Types of Redox Reactions

1. (A) **2.** (C) **3.** (A) **4.** (A)
5. (B) **6.** (B) **7.** (D) **8.** (C)
9. (D) **10.** (D) **11.** (B) **12.** (B)
13. (D) **14.** (D) **15.** (D) **16.** (A)
17. (A) **18.** (B)

Application of Redox Reaction

1. (B) **2.** (A) **3.** (C) **4.** (C)
5. (A)

Oxidation & Reduction

1. (D) **2.** (B) **3.** (D) **4.** (D)
5. (B) **6.** (C) **7.** (A) **8.** (C)
9. (C) **10.** (B) **11.** (C) **12.** (D)
13. (D)

Redox Reactions

1. (A) **2.** (D) **3.** (C) **4.** (B)
5. (D) **6.** (B) **7.** (B) **8.** (D)
9. (B) **10.** (C) **11.** (C) **12.** (D)

Concept of Equivalent and Titration

1. (D)	**2.** (D)	**3.** (A)	**4.** (C)
5. (B)	**6.** (B)	**7.** (D)	**8.** (C)
9. (A)	**10.** (B)	**11.** (A)	**12.** (D)
13. (A)	**14.** (D)	**15.** (B)	**16.** (D)
17. (C)	**18.** (B)	**19.** (C)	**20.** (A)
21. (B)	**22.** (C)		

EXERCISE-2

1. (D)	**2.** (C)	**3.** (B)	**4.** (D)
5. (D)	**6.** (A)	**7.** (A)	**8.** (C)
9. (C)	**10.** (D)	**11.** (D)	**12.** (B)
13. (B)	**14.** (A)	**15.** (A)	**16.** (D)
17. (A)	**18.** (A)	**19.** (D)	**20.** (B)
21. (B)	**22.** (D)	**23.** (D)	**24.** (B)
25. (B)	**26.** (D)	**27.** (C)	**28.** (B)
29. (B)	**30.** (D)	**31.** (A)	**32.** (D)
33. (C)	**34.** (D)	**35.** (D)	**36.** (D)
37. (C)	**38.** (A)	**39.** (B)	**40.** (B)
41. (B)	**42.** (A)	**43.** (B)	**44.** (A)
45. (A)	**46.** (B)	**47.** (C)	**48.** (C)
49. (C)	**50.** (B)	**51.** (D)	**52.** (A)
53. (C)	**54.** (B)	**55.** (B)	**56.** (A)
57. (C)	**58.** (C)	**59.** (B)	**60.** (C)
61. (B)	**62.** (C)	**63.** (A)	**64.** (D)
65. (B)	**66.** (B)		

NSEJS PYQ

1. (C)	**2.** (A)	**3.** (B)	**4.** (D)
5. (D)	**6.** (A)	**7.** (B)	**8.** (C)
9. (B)	**10.** (C)	**11.** (B)	**12.** (D)
13. (C)	**14.** (A)	**15.** (C)	**16.** (A)
17. (C)	**18.** (C)	**19.** (A)	**20.** (A)
21. (B)	**22.** (B)	**23.** (B)	**24.** (A)
25. (A)	**26.** (A)	**27.** (A)	**28.** (B)
29. (B, C)	**30.** (D)	**31.** (C)	**32.** (B)
33. (A, B, C)			

Advanced & Olympiad Challenger Questions

1. (D) **2.** (A) **3.** (C) **4.** (B)

5. (C) **6.** (C) **7.** (C) **8.** (C)

9. (A) **10.** (D) **11.** (D) **12.** (C)

13. (B) **14.** (D) **15.** (C) **16.** (A)

17. (i) 28 cm^3 (ii) 3

(iii) $[HNO_3] = 2.80 \times 10^{-3} \div 0.025 = 0.112$ mol dm^{-3}

18. $H_2C_2O_4$ = 0.9g (Oxalic acid)

$NaHC_2O_4$ = 1.12 g

19. i. Molar proportion $KBrO_3$: KBr = 1:5 ii. $1[KBrO_3] = 3[Br_2] = 3[I_2] = 6[Na_2S_2O_3]$

iii. 12ml of 0.05M $Na_2S_2O_3$ are required.

20. (I) $3NaOH + H_3PO_4 \; Na_3PO_4 + 3H_2O$ (II) 2.48 grams.

(III) 2.87 M (IV) 0.051 L

21. 41%

22. **(I) Sample (1)** 0.53g of $CaCO_3$, degree of hardness = 2.03ppm

Sample (2) 2.5g of $CaCO_3$, degree of hardness = 2.5ppm

(II) $Ca(HCO_3)_2 + 2NaOH \longrightarrow CaCO_3 + Na_2CO_3 + 2H_2O$

(Any one reaction either with Calcium or Magnesium)

$CaSO_4 + Na_2CO_3 \longrightarrow CaCO_3 + Na_2SO_4$

$2MgCl_2 + 2Na_2CO_3 \longrightarrow 2MgCO_3 + 4NaCl$

(Any two reactions either with sulphate or chloride of Calcium or Magnesium)

23. (A) $2KMnO_4 + 5H_2C_2O_4 + 3H_2SO_4 \longrightarrow 1K_2SO_4 + 2MnSO_4 + 8H_2O + 10CO_2$

(B) (i) $KMnO_4$ is an oxidizing agent as it is getting reduced to $MnSO_4$.(+7 to +2)

(ii) $H_2C_2O_4$ is a reducing agent, as it is getting oxidised to CO_2 (+3 to +4)

(C) 4.45×10^{-3} moles

(D) 0.445 gm

(E) 28.91%